SUSTech Southern University of Science and Technology

SME206, Signals and Systems

Lab1 – Matlab Intro & Signals Basic

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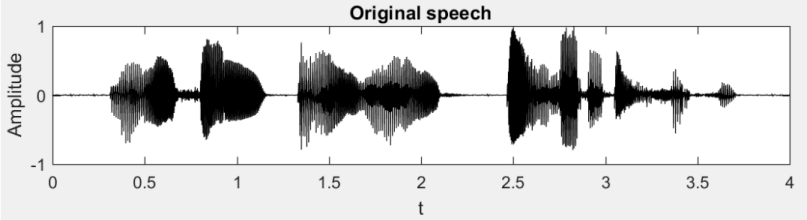
Outline

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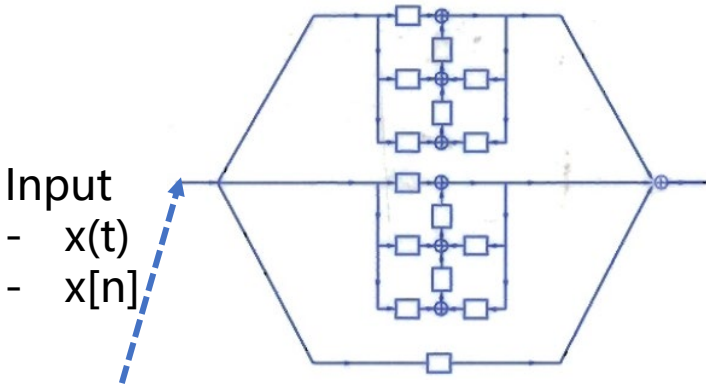
- **Lab Intro**
- **Matlab Intro**
- Matlab Syntax
- Signals Basic



Lab Intro / Background



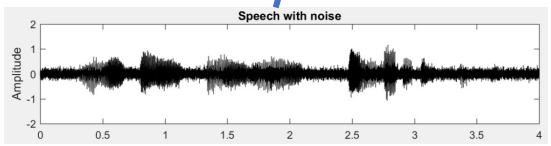


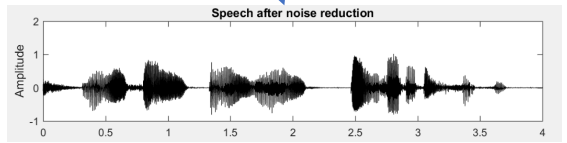


Input
- x(t)
- x[n]

Output
- y(t)
- y[n]

i.e. Filter for speech denoising





Signals

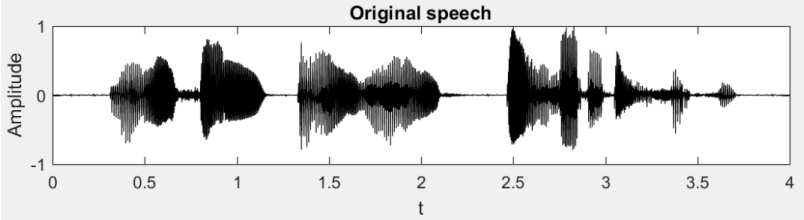
- A function of one or more variables, contains information on the properties of certain phenomena.
- Continuous time signals (i.e. speech) vs Discrete time signals (i.e. stock)
- Periodic vs Non Periodic

Systems

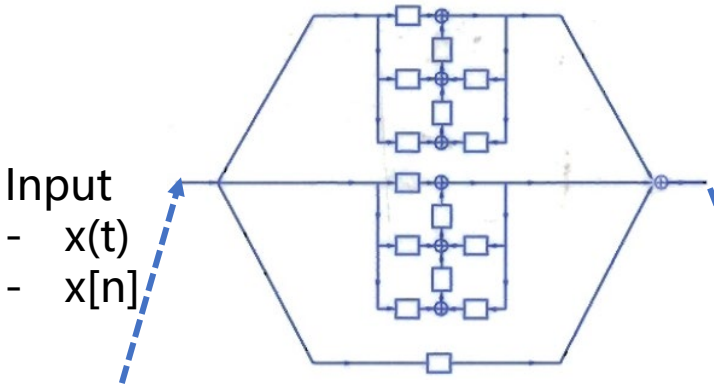
- To respond to given input signals, thereby generating additional signals, or producing certain desired properties.
- Highlight: Linear Time Invariant System

Lab Intro / Background

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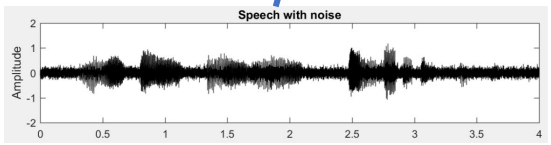


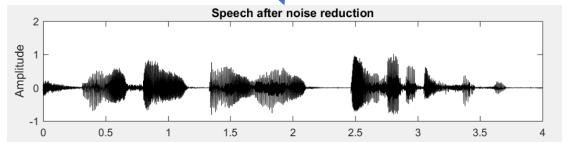


Input
- x(t)
- x[n]

Output
- y(t)
- y[n]

i.e. Filter for speech denoising





Signals： **Continuous /Discrete x Periodic/Non-Periodic**

Systems： **Linear Time Invariant System**



using **Matlab** for Signals and System Analysis

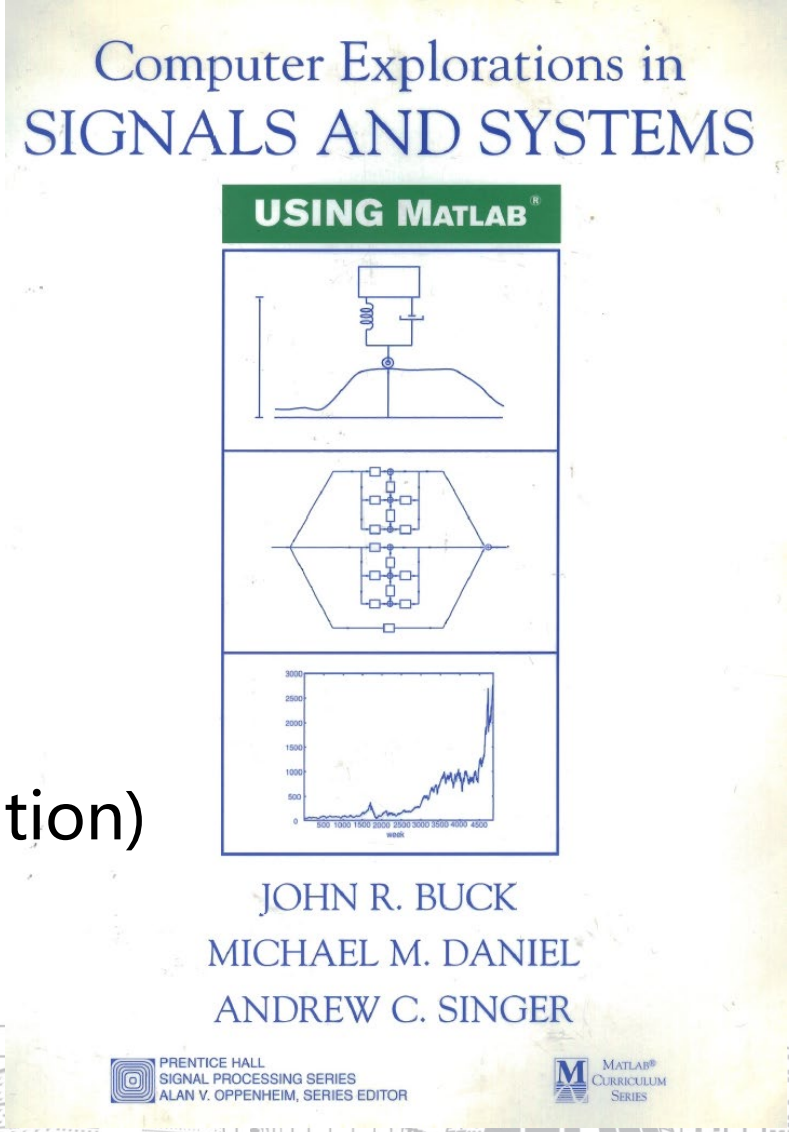
⇒ Deepen understanding: Visualize

⇒ Theoretical practice: Tools for problem analysis and solving

Lab Intro / Content

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- Lab1：Matlab Intro & Signals Basic，2 Weeks
- Lab2：Linear Time Invariant System，2 Weeks
- Lab3：Fourier Series of Periodic Signals，2 Weeks
- Lab4：Fourier Transform，3 Weeks
- Project：TBD，5 Weeks (Last 2 weeks for presentation)



Lab Intro / Evaluation（Lab Part)

- Overall proportion：Lab account for 30% of total score => 30 points
- Attendance，5 points (including lecture)
- Lab Report x4，10 points
  - Individual work
  - 2.5 points / report (学号_姓名_labN.pdf)
- Project x1，15 points
  - Teamwork, 2 members per group
  - Report (学号_姓名_Project.pdf) + Presentation

Submit report to Blackboard on time.

Note：if report submitted 1 day late, the score will only be 50%. No points will be rewards if too late.

Showcase: Using Matlab For Signals Analysis

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全国大学生电子设计竞赛，2023年国赛H题

信号分离装置（H 题）

【本科组】

一、任务

设计并制作信号分离装置，如图 1 所示。一台双路输出信号源输出 2 路周期信号 A 和 B（频率范围：20kHz ~100kHz，且 $f_A < f_B$；峰峰值均为 1V），经增益为 1 的加法器产生混合信号 C，信号 C 通过分离电路分离出信号 A′和 B′。要求信号 A′和 B′相比信号 A 和 B 波形无失真，A′和 A、B′和 B 的波形在示波器上能连续稳定同频显示。

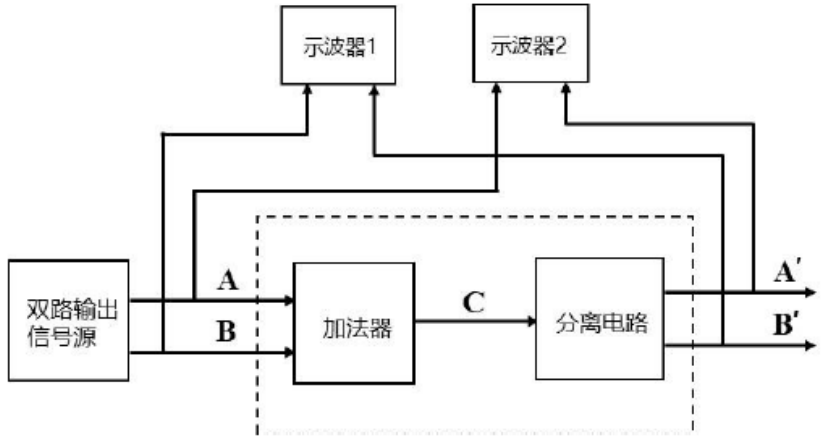


图 1 信号分离装置框图

1．基本要求

（1）制作增益为 1 的加法器，实现 C=A+B。

（2）信号 A 和 B 均为正弦波，f_A=50kHz，f_B=100kHz。要求装置能正确分离出信号 A′和 B′，且峰峰值均不小于 1V。

（3）信号 A 和 B 均为正弦波，频率分别为 10kHz 的整数倍。要求装置能正确分离出信号 A′和 B′，且峰峰值均不小于 1V。

2．发挥部分

（1）信号 A 和 B 分别为正弦波或三角波，频率分别为 5kHz 的整数倍。要求装置能正确分离出信号 A′和 B′，且峰峰值均不小于 1V。

（2）发挥部分（1）中，信号 A 和 B 均为正弦波，且 f_B 是 f_A 的整数倍。要求装置能设置并控制信号 B′与 A′的初相位差，范围 0° ~180° ，设置分辨率 5° ，误差绝对值不大于 5° 。

（3）其他。

Showcase: Using Matlab For Signals Analysis

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全国大学生电子设计竞赛，2023年国赛H题

Input
A: sine wave, 10kHz
B: sine wave, 30kHz

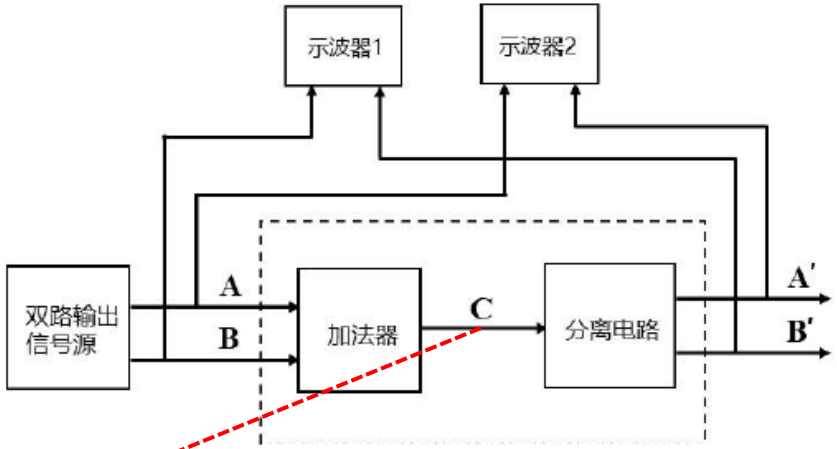
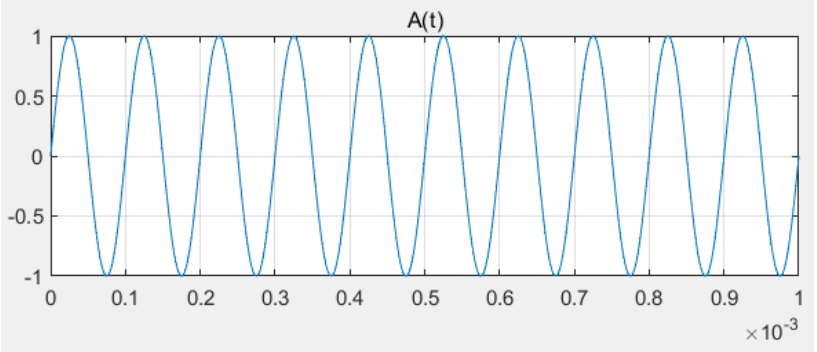
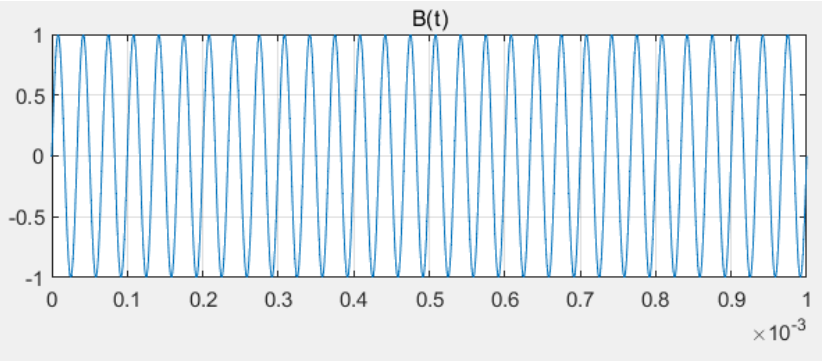
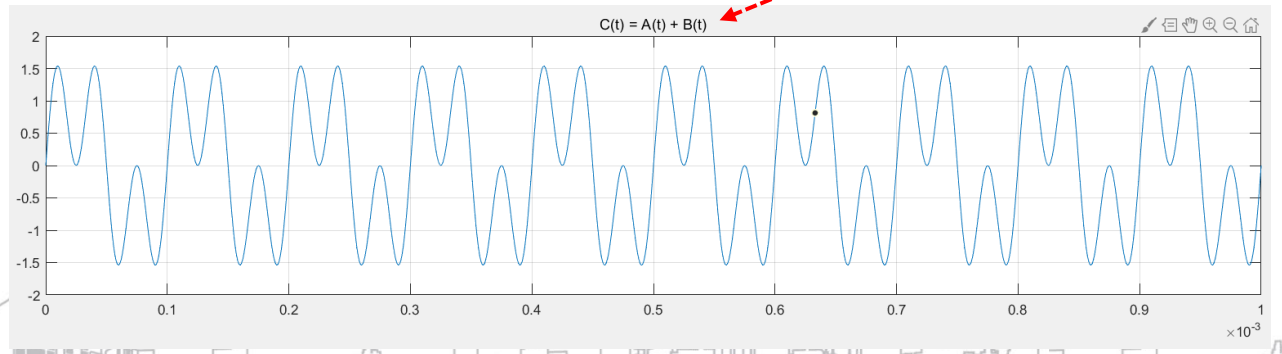


图 1 信号分离装置框图

Sampling







Showcase: Using Matlab For Signals Analysis

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全国大学生电子设计竞赛，2023年国赛H题

Matlab helps with theoretical Analysis and Validation

- Spectrum of C under difference input A and B
- Reasonable sampling rate for C ?
- How about adding noise to C ?

Input
A: sine wave, 10kHz
B: sine wave, 30kHz

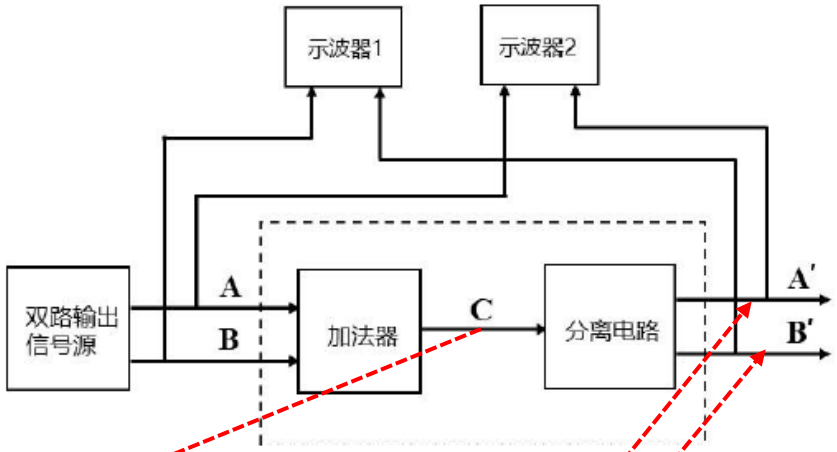
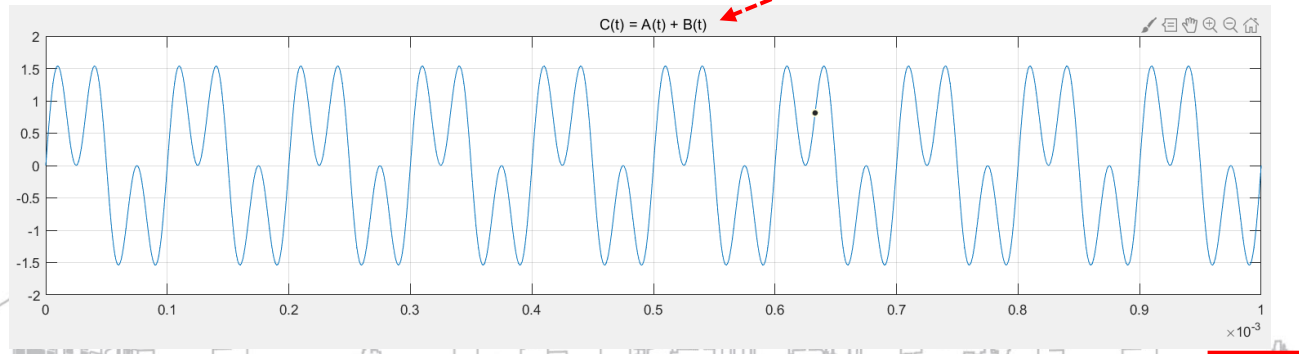


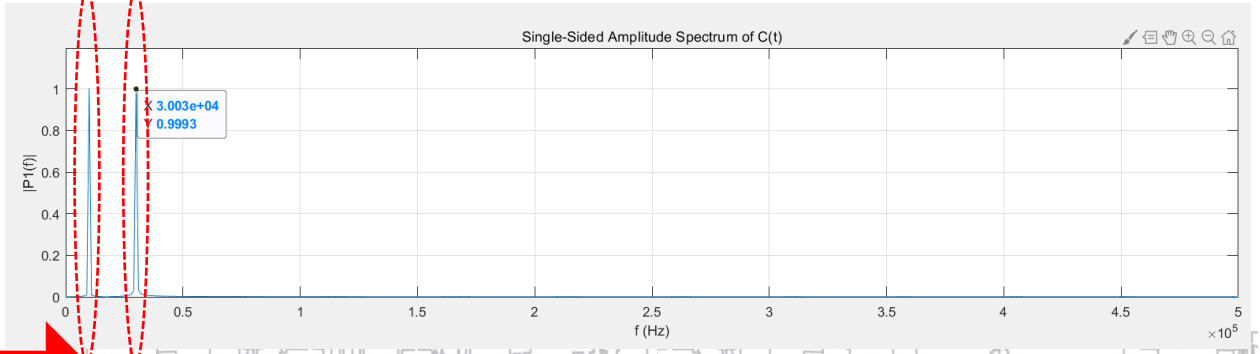
图 1 信号分离装置框图

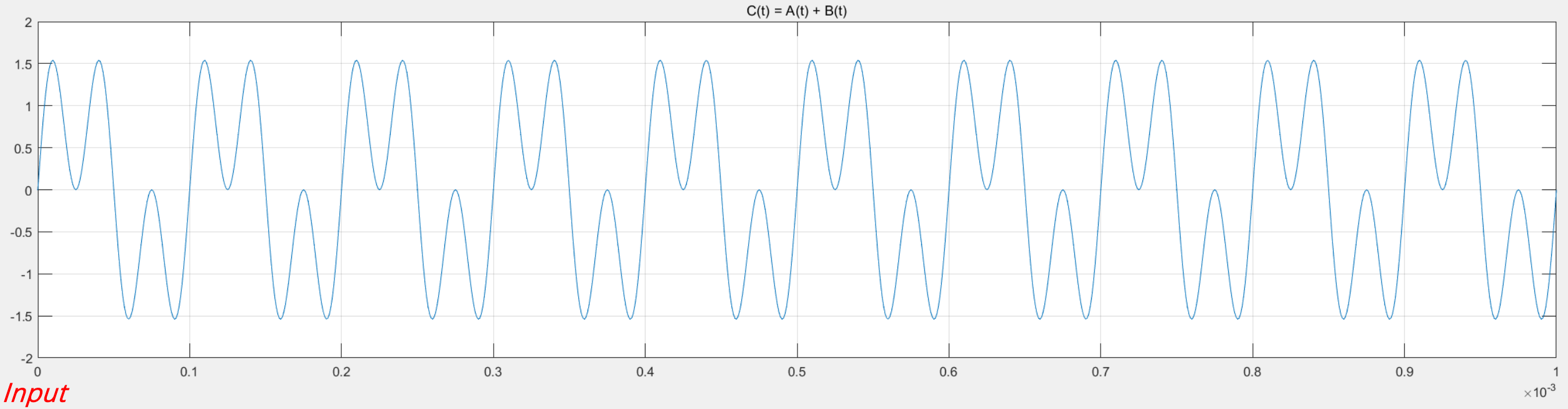
Sampling

Output - Signal separation

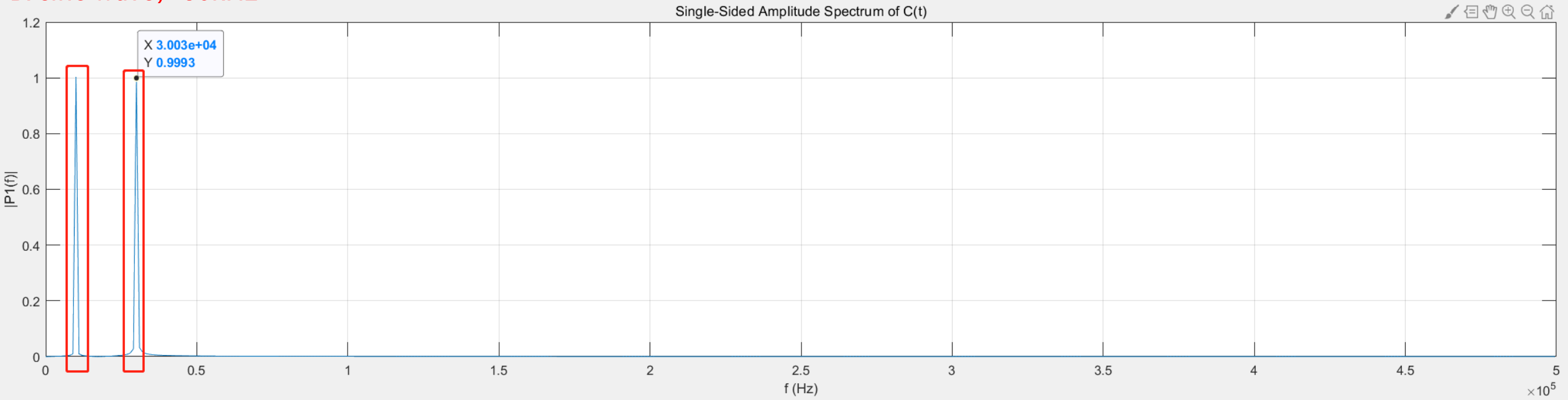


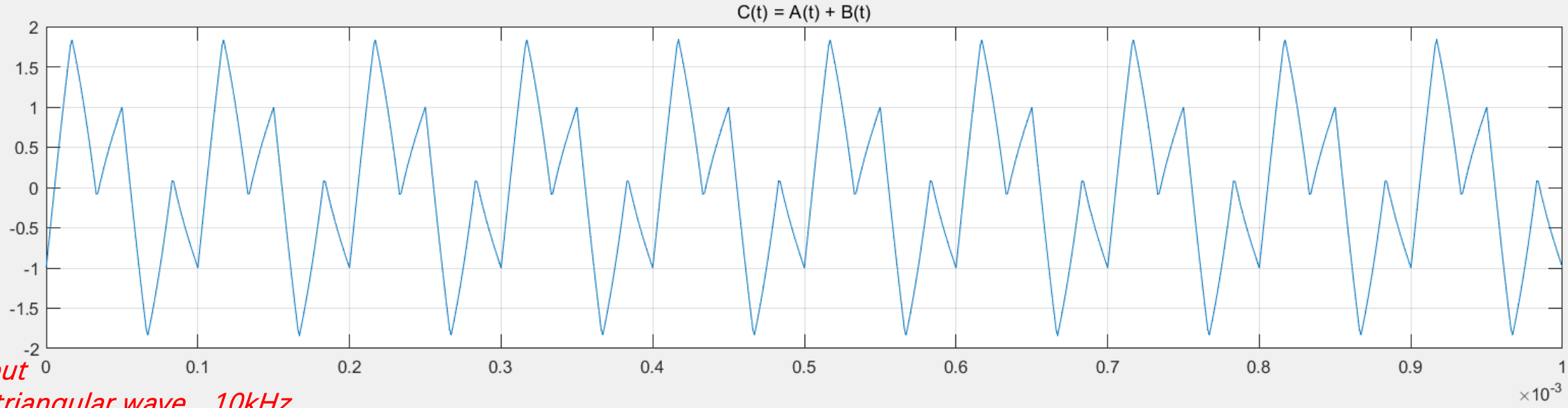
FFT



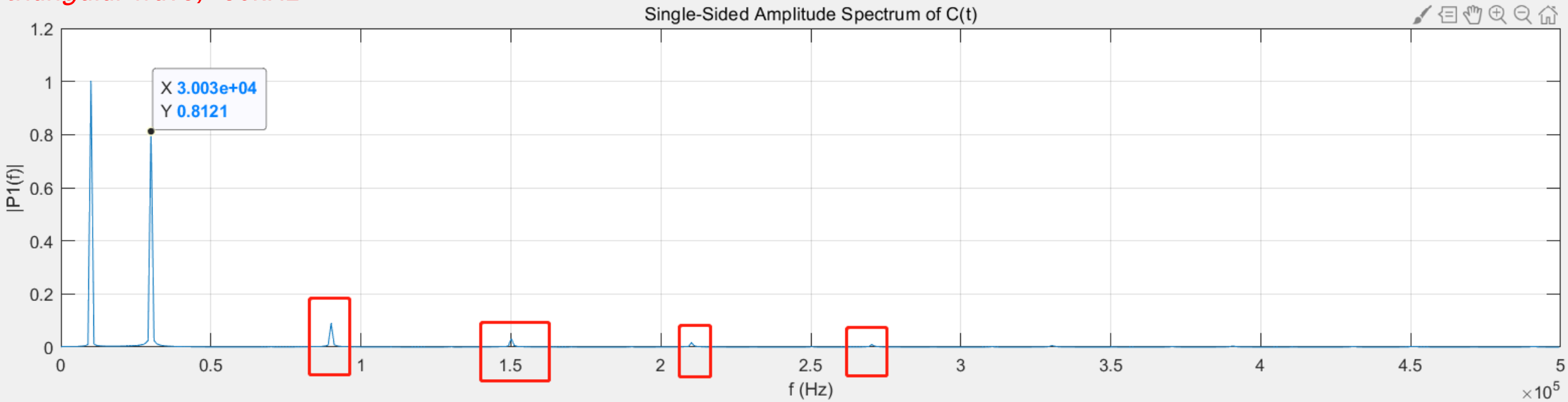


Input
A: sine wave, 10kHz
B: sine wave, 30kHz





Input
A: triangular wave, 10kHz
B: triangular wave, 30kHz

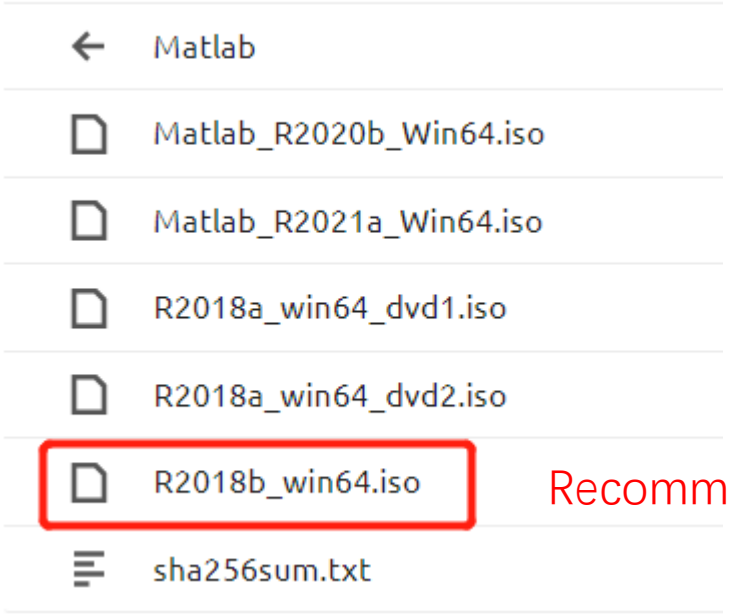


Matlab Intro / Download

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Downloads from SUSTech network

https://dl.cra.moe/download/Matlab/Windows/



Recommend

Matlab Intro / Installation & Register

School library resources

https://lib.sustech.edu.cn/gjyrj_116/list.htm

商业数学软件——MATLAB（数学系采购）

MATLAB 是美国MathWorks公司出品的商业数学软件，用于算法开发、数据可视化、数据分析以及数
括MATLAB和Simulink两大部分。数学系已购买Matlab软件全模块功能 服务期延长至2025年12月，

请依照安装激活指南操作：

MATLAB安装指南.pdf

MATLAB安装指南视频（建议观看）.mp4

MATLAB安装文件网盘链接 （提取码：7fxr）

MATLAB Campus-Wide License更新指南.docx

MATLAB学习&教学资源.rar

培训短视频及热门视频.rar

Link a License to my MathWorks Account (huangjw3@mail.sustech.edu.cn)

License Number* or Activation Key

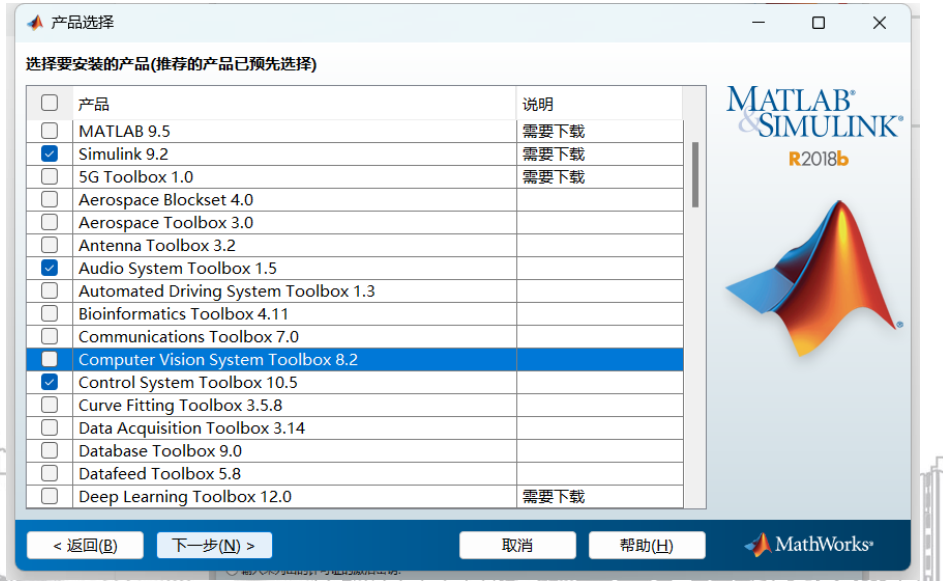
Link License

Obtain your license number by running MATLAB and typing 'license' at the command prompt or contact your MATLAB administrator.

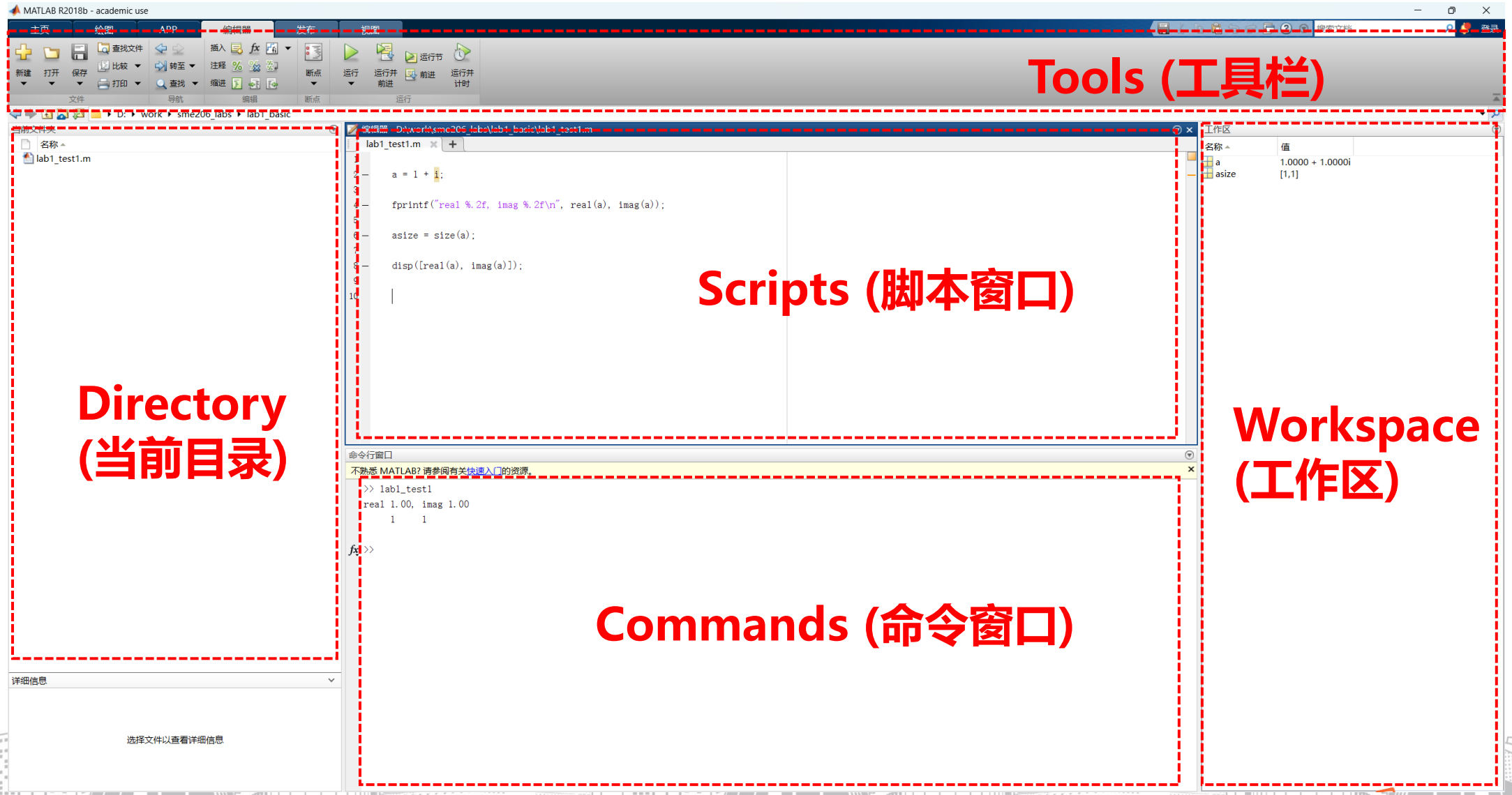
**Some Licenses may require you to enter an Activation Key.*

注意：

- 中间需要采用 mail.sustech.edu.cn 邮箱注册，并绑定学生的 activation key
- 本课程用到的 toolbox： audio system toolbox, signal processing toolbox, control system toolbox, symbolic math toolbox

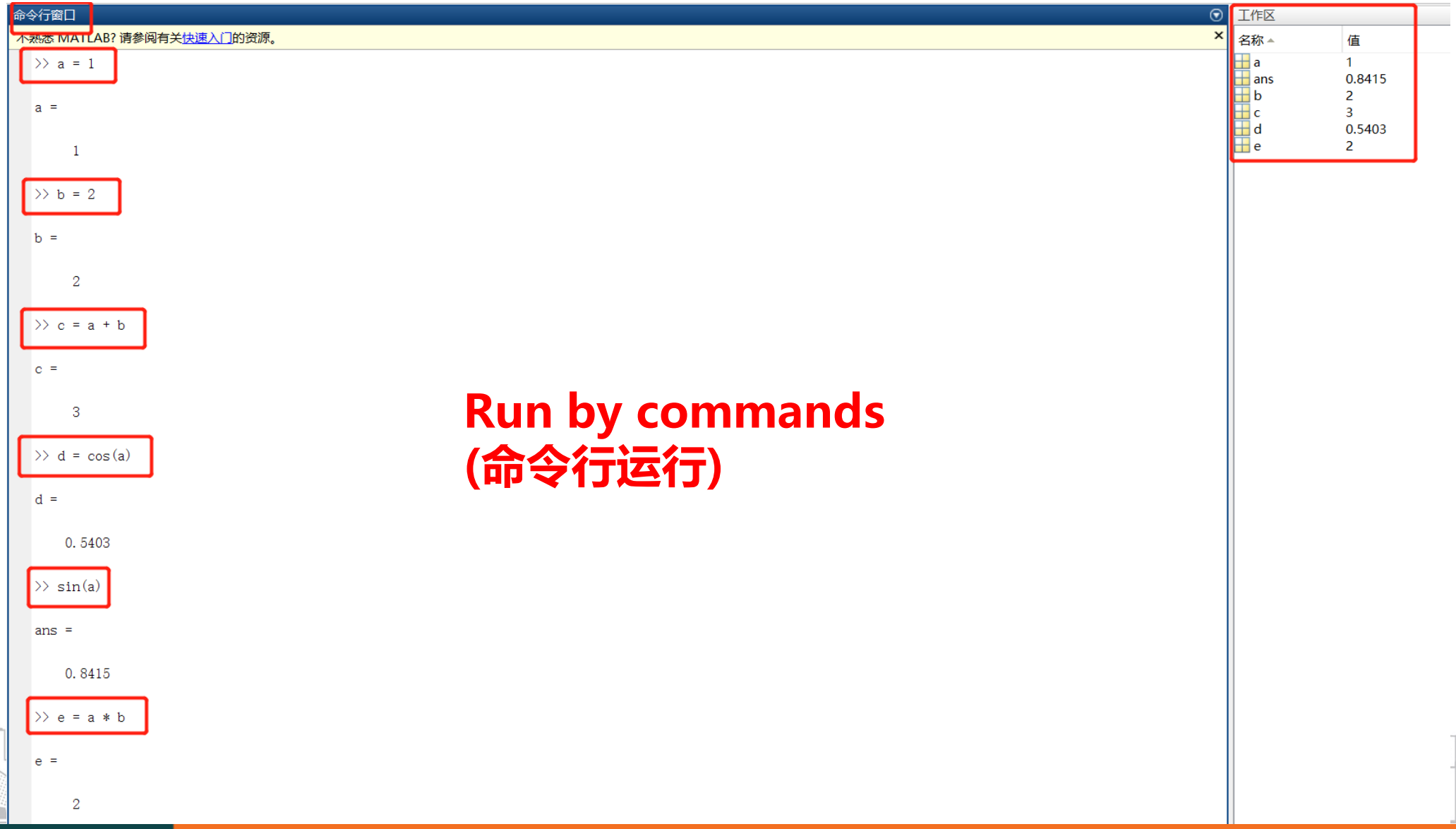


Matlab Intro / UI



Matlab Intro / HelloWorld

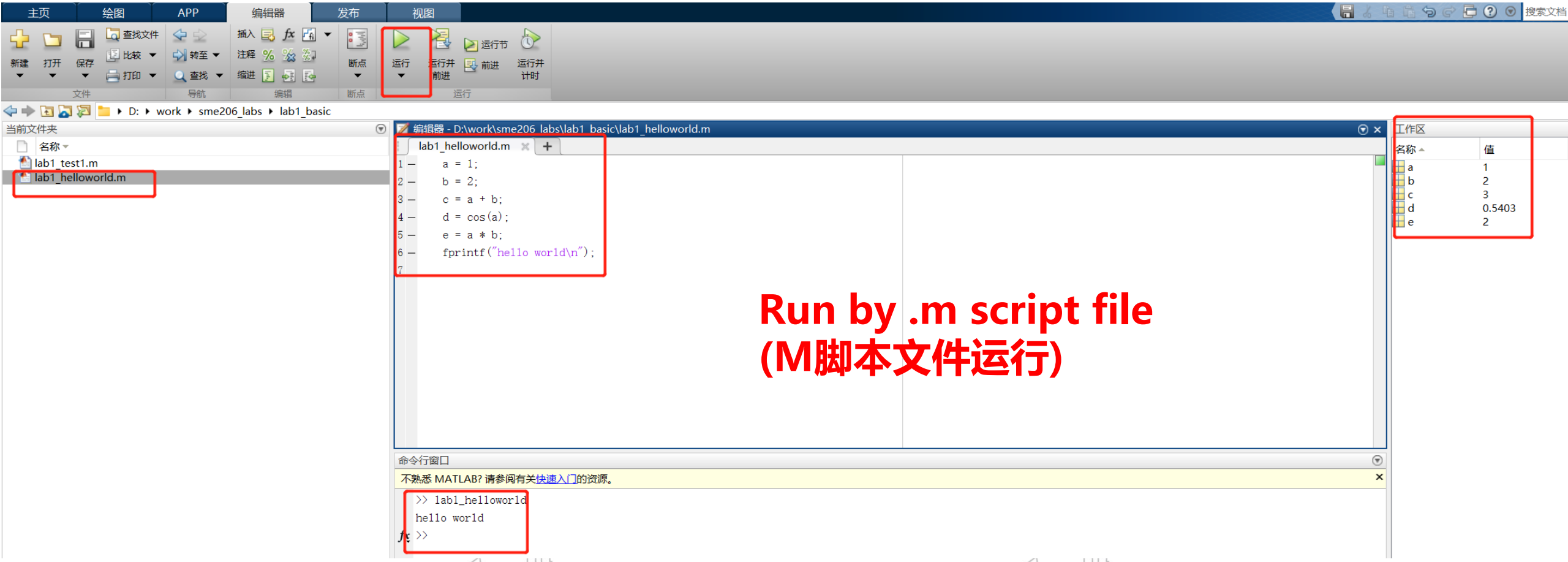
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Run by commands
(命令行运行)

Matlab Intro / HelloWorld

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Run by .m script file
(M脚本文件运行)

More：Matlab Commands

- help [Function name]: function usage (built-in)
- lookfor [Keywords]： search for help documents
- doc [Function name]： function usage (more details)

```
>> help cos
cos - 以弧度为单位的参数的余弦

    此 MATLAB 函数 返回 X 的每个元素的余弦。cos
    [-1 ,1] 内的实数值。对于 X 的复数值，cos 返

    Y = cos(X)

    另请参阅 acos, acosd, cosd, cosh, cospi
```

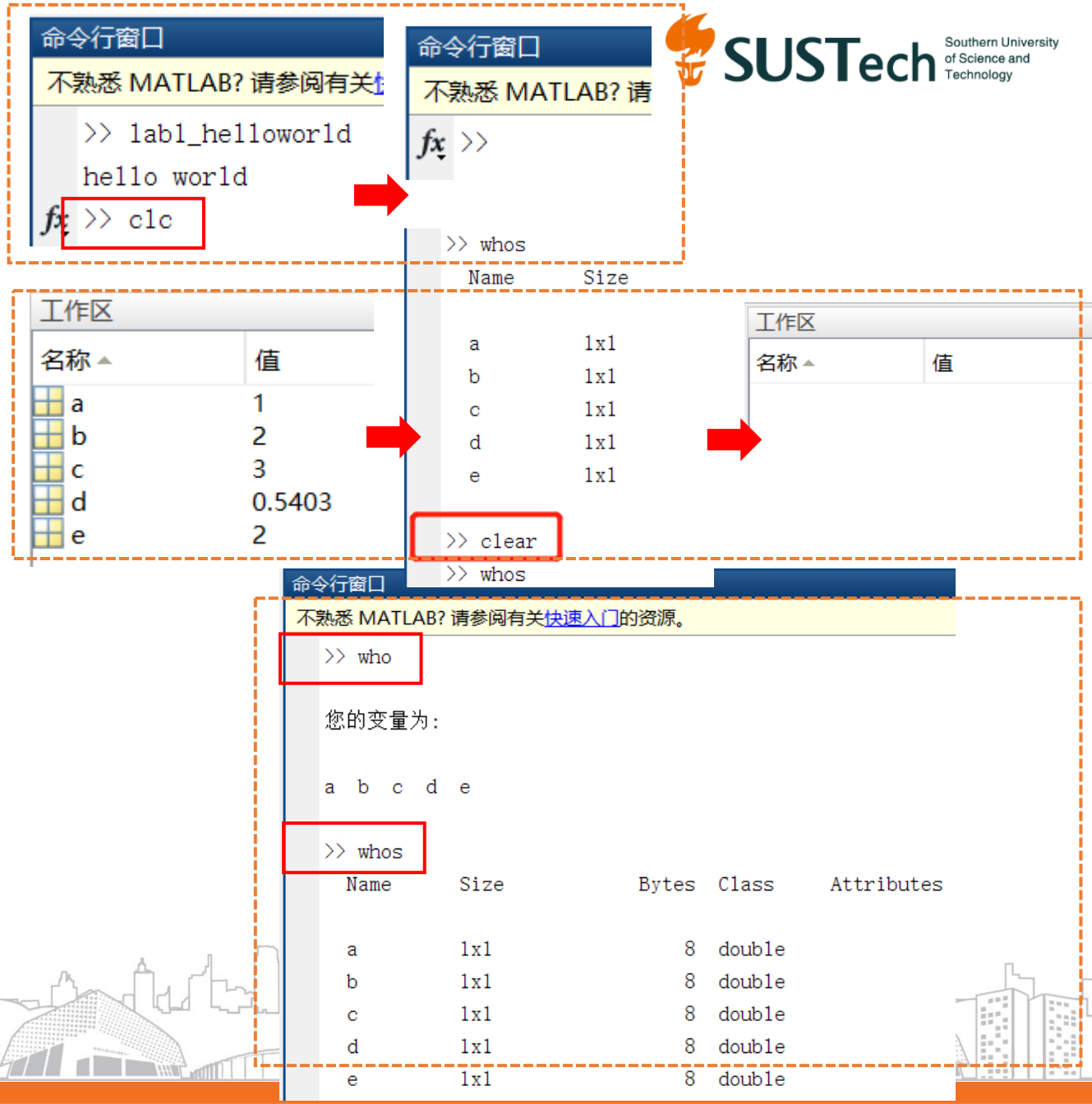
```
>> doc sin
```




```
>> lookfor sin
detrend          - Remove a linear trend from a vector, usually for FFT processing.
fillmissing      - Fill missing entries
ismissing        - Find missing entries
rmmissing        - Remove rows or columns with missing entries
standardizeMissing - Convert to standard missing data
cell2mat         - Convert the contents of a cell array into a single matrix.
isfloat          - True for floating point arrays, both single and double.
```

More：Matlab Commands

- clc : clear up command window
  (清空命令窗口的显示内容)
- clear： delete variables in workspace
  (删除工作空间的变量)
- who： list variables in workspace
  (列出工作区的变量)
- whos：list more details of variables …
  (列出工作区的变量、大小及类型)



Class Task：Matlab experience

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1）Run the following code by the **Command line (命令行)**

```
syms x;
syms y;
z = sym(17*x^2-16*abs(x)*y+17*y^2-225);
ezplot(z)
```

- Observation：Output figure and workspace
- More：Find out the usage / meanings of ‘syms’, ‘sym’, ‘ezplot’ by ‘help’ command



Class Task：Matlab experience

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2）Run the following code by the **.m script file (M脚本)**

```
clear; clc;
[x, y] = meshgrid(-10:0.01:10);
z = -(17 * x .^2 - 16 * y .* abs(x) + 17 * y .^2);
[c,h] = contourf(z, 100);
set(h,'linestyle','none');
```

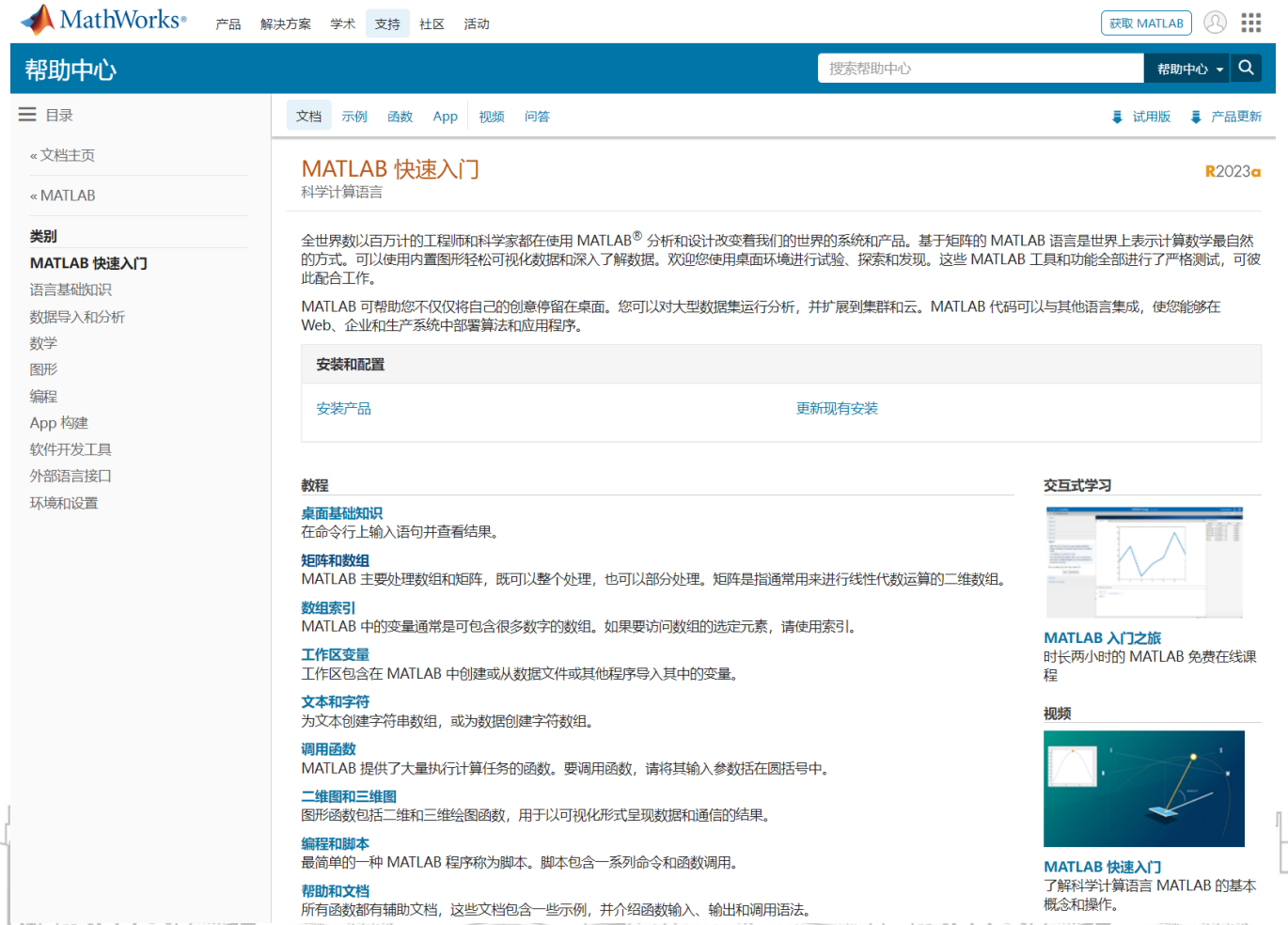
- Observation：Output figure and workspace
- More：Find out the usage / meanings of unknown function by ‘help’ command
- Extension：how about adjust z and other variables, to obtain more interesting figure ?



More：Matlab official guide

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https://ww2.mathworks.cn/help/matlab/getting-started-with-matlab.html



撸起袖子加油干

Practice make perfect

Outline

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- Lab Intro
- Matlab Intro
- **Matlab Syntax**
- Signals Basic



Matlab Syntax / Outline

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- Data & Variables (数据&变量)
- Operator (运算符)
- Function (函数)
- Flow Control (流程控制)
- Plot (画图)



Matlab Syntax / Data & Variables

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Variables (变量)

- Basic element of any programming language
- Data is stored in memory in the form of varibles
- Unlike C / Java，Matlab **not require** variable declaration (变量声明) or specifying variable types (变量类型)，can simply assign values (similar to Python)

Naming Rules (命名规则)

- Case sensitivity，‘aa’ and ‘AA’ are different variables
- Length of name <= 63 characters （for 64 bits systems）
- Name must start with a letter，any combination of letters (a-z A-Z), number (0-9) , and underscores(_)
- DON’T use grammer keywords （checked by ’iskeyword’）

```
>> iskeyword case

ans =

  logical

    1
```

Matlab Syntax / Data & Variables

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Data Type (数据类型)

- Matrix(矩阵): **Most basic data type**， an array essentially
- Vector (向量)： A matrix of 1 row (1xN) or 1 column (Nx1)
- Number： 1x1 matrix
- Cell (元胞数组) : a multidimensional matrix of elements in any form of array. (任意形式的数组为元素的多维矩阵). e.g. {90.0, “hello”}
- Structure (结构体数组): Similar to Cell, its element has specific attribute names

Matlab Syntax / Data & Variables

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Constants: Pre-defined variables

Name	Description
ans	Default variable for the result
pi	π
inf	Infinite, e.g. 1/0
NaN	Not A Number, e.g. 0/0, inf/inf, 0*inf
i, j	Imaginary unit (虚数单位)

```
>> pi

ans =

    3.1416

>> a = 1/0

a =

   Inf

>> b = 0/0

b =

   NaN
```

```
>> c = 1 + j

c =

   1.0000 + 1.0000i
```

Matlab Syntax / Data & Variables

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Real Number

- Decimal： -2, 1.23
- Scientific computing method： 1.23e4

```
>> a = -2

a =

    -2
```

```
>> b = 1.23

b =

    1.2300
```

```
>> c = 1.23e4

c =

       12300
```

Matlab Syntax / Data & Variables

Real Number

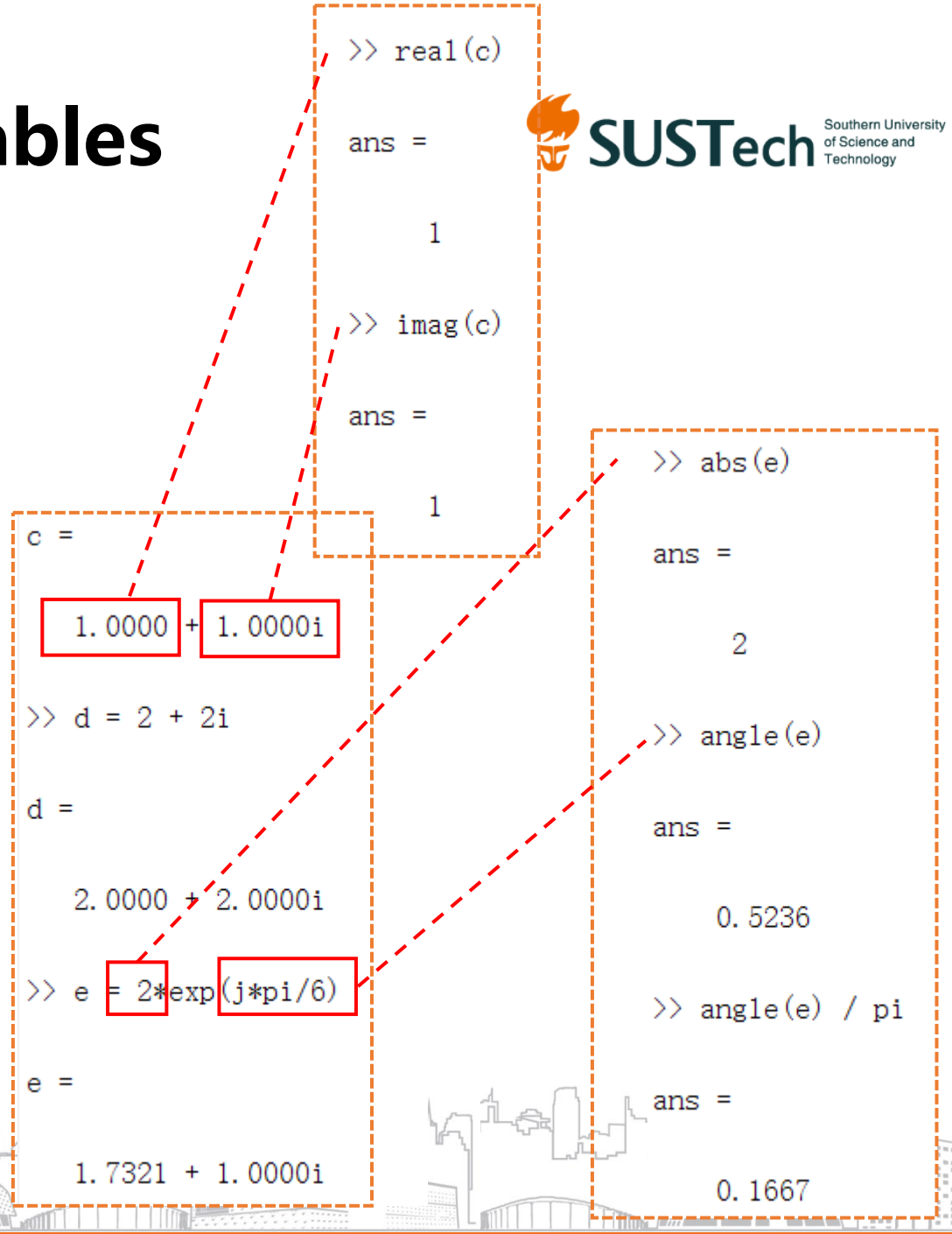
- Decimal： -2, 1.23
- Scientific computing method： 1.23e4

Complex Number

- z = a+bj, or z = a+bi, or z = r*exp(i*θ)

Functions for complex number:

- a = real(z) % Real Part
- b = imag(z) % Imaginary part
- r = abs(z) % amplitude
- theta = angle(z) % phase angle



Matlab Syntax / Data & Variables

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Real Number

- Decimal： -2, 1.23
- Scientific computing method： 1.23e4

Complex Number

- z = a+bj, or z = a+bi, or z = r*exp(i*θ)

Data type of matrix elements

- Integer (整型)
  - Unsigned (无符号, uint8/uint16/uint32/uint64),
  - Signed (符号型, int8/…/int64)
- Float (浮点型)
  - 单精度 (32 bit)，双精度 (64 bit, 默认)
- Logical (逻辑型)
- Character (字符型)

```
>> iskeyword ll

ans =

  logical

   0

>> logical(1)

ans =

  logical

   1

>> s = "string"
>> s

s =

    "string"
```

Matlab Syntax / Data & Variables

SUSTech Southern University of Science and Technology

Input for Matrix (矩阵输入)

- Explicit element list input
  - Overall：enclosed in square brackets [ ] (方括号)
  - In line (行内)：separated by commas (,) or space
  - Interline (行间)：separated by semicolons (; 分号) or carriage returns (回车)
  - Element：numerical value or expression

- Generate by statements
  - from:step:to (step，default 1 if missing)
  - …

```
>> m1 = [1, 2; 3, 4]

m1 =

     1     2
     3     4

>> m2 = [1 2
3 4
]

m2 =

     1     2
     3     4
```

```
>> m3 = 1:10

m3 =

     1     2     3     4     5     6     7     8     9    10

>> m4 = 1:2:10

m4 =

     1     3     5     7     9
```

Matlab Syntax / Data & Variables

Input for Matrix (矩阵输入)

- Explicit element list input
  - Overall： enclosed in square brackets [ ] (方括号)
  - In line (行内)： separated by commas (,) or space
  - Interline (行间)： separated by semicolons (; 分号) or carriage returns (回车)
  - Element： numerical value or expression

- Generate by statements
  - from:step:to (step， default 1 if missing)
  - linspace(a, b, n) – linear n-division (线性n等分) between a and b
  - Other functions
    - zeros(m, n)
    - ones(m, n)
    - rand(m, n) – (0,1) uniform distribution
    - randn(m, n) – normal distribution
    - eye(m, n)
    - …

```
>> linspace(0, 10, 4)

ans =

         0    3.3333    6.6667   10.0000
```

```
>> zeros(4, 4)

ans =

     0     0     0     0
     0     0     0     0
     0     0     0     0
     0     0     0     0
```

```
>> ones(4, 4)

ans =

     1     1     1     1
     1     1     1     1
     1     1     1     1
     1     1     1     1
```

```
>> randn(2, 3)

ans =

    1.4090    0.6715    0.7172
    1.4172   -1.2075    1.6302

>> rand(3, 2)

ans =

    0.6787    0.3922
    0.7577    0.6555
    0.7431    0.1712
```

```
>> eye(2,2)

ans =

     1     0
     0     1
```

Matlab Syntax / Data & Variables

SUSTech Southern University of Science and Technology

Index of matrix elements

- Full index method (全下标方式)：A(i, j) – in i-th row and j-th column of A
  - **Note：index start from 1**

- Single index method (单下标方式)
  - Take A (m x n) as a 1-dimensional column and concatenate all the columns of the matrix A from left to right (把 m x n 的 A 看出1维长列，把矩阵的所有列从左到右依次连起来)
  - **Optional**

```
>> m1 = [1, 2; 3, 4]

m1 =

   1   2
   3   4

>> m1(1, 1)

ans =

   1

>> m1(2, 1)

ans =

   3
```

Recommend way

```
>> m1(3)

ans =

   2
```

m1(3) = m1(1, 2)

Matlab Syntax / Data & Variables

Extraction of Matrix (矩阵的抽取)

- Use matrix subscripts to generate submatrices,

such as:

```
>> A = [1, 2, 3; 4, 5, 6; 7, 8, 9]

A =

   1   2   3
   4   5   6
   7   8   9
```

⇒ In 1st~2nd row，and 2nd~3rd column

```
>> A(1:2, 2:3)

ans =

   2   3
   5   6
```

⇒ In 1st、3nd rows，and 1nd、3rd columns

```
>> A([1,3], [1, 3])

ans =

   1   3
   7   9
```

⇒ In all rows，and 3rd column

```
>> A(:,3)

ans =

   3
   6
   9
```

⇒ In 2nd ~ the last (end) row，and the last (end) column

```
>> A(2:end, end)

ans =

   6
   9
```

⇒ Take elements with single index of 9, 8, 7, and 6 to form a submatrix

```
>> A([9, 8; 7, 6])

ans =

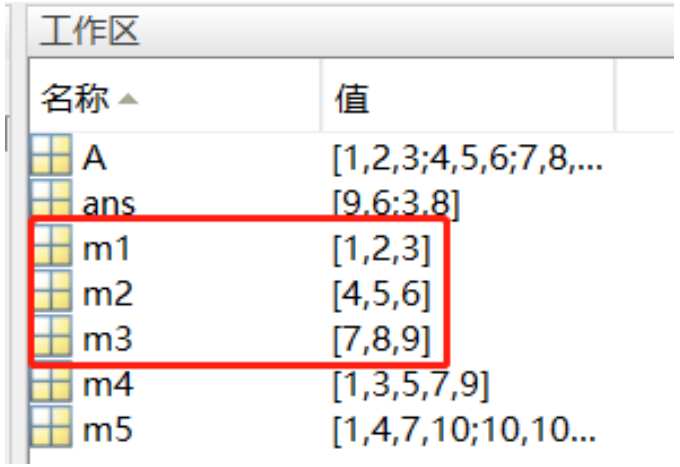
   9   6
   3   8
```

Matlab Syntax / Data & Variables

SUSTech Southern University of Science and Technology

Matrix Merging (矩阵的合并)

- Using [] to merge small matrices into a large one.



⇒ [m1, m2, m3]: column connection (列联结)

⇒ [m1; m2; m3]: Row connection (行联结)

```
>> [m1, m2, m3]

ans =

     1     2     3     4     5     6     7     8     9
```

```
>> [m1; m2; m3]

ans =

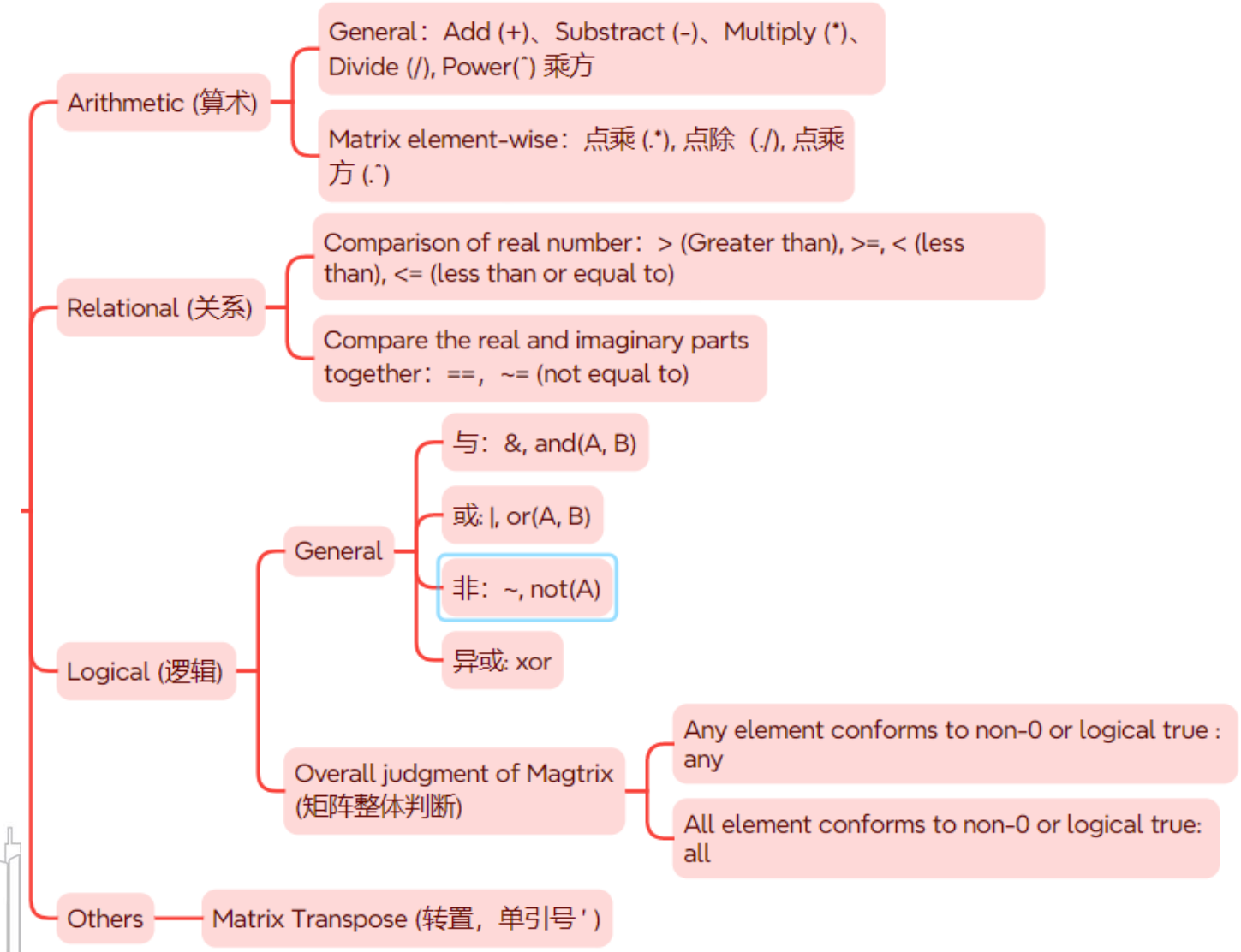
     1     2     3
     4     5     6
     7     8     9
```

Matlab Syntax / Outline

- Data & Variables (数据&变量)
  - Data type，basic data type
  - Variables：Naming rules，Pre-defined variables
  - Real & Complex number
  - Input for matrix
  - Matrix indexing, extraction and merging
- **Operator (运算符)**
- Function (函数)
- Flow Control (流程控制)
- Plot (画图)



Matlab Syntax / Operator



Matlab Syntax / Operator

SUSTech Southern University of Science and Technology

- Arithmetic (算术)
  - General：Add (+)、Substract (-)、Multiply (*)、Divide (/), Power(^) 乘方
  - Matrix element-wise：点乘 (.*), 点除 (./), 点乘方 (.^)
- Relational (关系)
  - Comparison of real number：> (Greater than), >=, < (less than), <= (less than or equal to)
  - Compare the real and imaginary parts together：==, ~= (not equal to)
- Logical (逻辑)
  - General
    - 与：&, and(A, B)
    - 或: |, or(A, B)
    - 非：~, not(A)
    - 异或: xor
  - Overall judgment of Magtrix (矩阵整体判断)
    - Any element conforms to non-0 or logical true : any
    - All element conforms to non-0 or logical true: all
- Others
  - Matrix Transpose (转置，单引号 ')

```
A =

     1     2     3
     4     5     6
     7     8     9
```

```
B =

     1     1     1
     2     2     2
     3     3     3
```

```
>> A + B

ans =

     2     3     4
     6     7     8
    10    11    12
```

```
>> A .* B

ans =

     1     2     3
     8    10    12
    21    24    27
```

```
>> A * B

ans =

    14    14    14
    32    32    32
    50    50    50
```

```
>> A ./ B

ans =

    1.0000    2.0000    3.0000
    2.0000    2.5000    3.0000
    2.3333    2.6667    3.0000
```

```
>> B / A
警告: 矩阵接近奇异值，或者缩放错误。

ans =

   -0.3750    0.4167   -0.0417
   -0.7500    0.8333   -0.0833
   -1.3000    1.6000   -0.3000
```

Matlab Syntax / Operator

SUSTech Southern University of Science and Technology

- Arithmetic (算术)
  - General：Add (+)、Substract (-)、Multiply (*)、Divide (/), Power(^) 乘方
  - Matrix element-wise：点乘 (.*), 点除 (./), 点乘方 (.^)
- Relational (关系)
  - Comparison of real number：> (Greater than), >=, < (less than), <= (less than or equal to)
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- Logical (逻辑)
  - General
    - 与：&, and(A, B)
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    - 非：~, not(A)
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  - Overall judgment of Magtrix (矩阵整体判断)
    - Any element conforms to non-0 or logical true : any
    - All element conforms to non-0 or logical true: all
- Others
  - Matrix Transpose (转置，单引号 ')

```
A =

     1     2     3
     4     5     6
     7     8     9
```

```
>> A >= 3

ans =

  3×3 logical 数组

   0   0   1
   1   1   1
   1   1   1
```

```
>> A == (1 + i)

ans =

  3×3 logical 数组

   0   0   0
   0   0   0
   0   0   0

>> A ~= 1

ans =

  3×3 logical 数组

   0   1   1
   1   1   1
   1   1   1
```

Matlab Syntax / Operator

SUSTech Southern University of Science and Technology

```
A =
     1     2     3
     4     5     6
     7     8     9

C =
     1     1     1
     0     0     0
     1     1     1
```

- Arithmetic (算术)
  - General：Add (+)、Substract (-)、Multiply (*)、Divide (/), Power(^) 乘方
  - Matrix element-wise：点乘 (.*), 点除 (./), 点乘方 (.^)
- Relational (关系)
  - Comparison of real number：> (Greater than), >=, < (less than), <= (less than or equal to)
  - Compare the real and imaginary parts together：==, ~= (not equal to)
- Logical (逻辑)
  - General
    - 与：&, and(A, B)
    - 或: |, or(A, B)
    - 非：~, not(A)
    - 异或: xor
  - Overall judgment of Magtrix (矩阵整体判断)
    - Any element conforms to non-0 or logical true : any
    - All element conforms to non-0 or logical true: all
- Others
  - Matrix Transpose (转置，单引号 ')

```
>> A & C

ans =

  3×3 logical 数组

   1   1   1
   0   0   0
   1   1   1

>> A | C

ans =

  3×3 logical 数组

   1   1   1
   1   1   1
   1   1   1
```

```
>> xor(A, C)

ans =

  3×3 logical 数组

   0   0   0
   1   1   1
   0   0   0
```

```
>> all(C)

ans =

  1×3 logical 数组

   0   0   0
```

```
>> all(C, 'all')

ans =

  logical

   0
```

```
>> any(C, 1)

ans =

  1×3 logical 数组

   1   1   1
```

```
>> any(C, 2)

ans =

  3×1 logical 数组

   1
   0
   1
```

```
>> A'

ans =

     1     4     7
     2     5     8
     3     6     9
```

Matlab Syntax / Outline

SUSTech Southern University of Science and Technology

- Data & Variables (数据&变量)
- Operator (运算符)
- **Function (函数)**
- Flow Control (流程控制)
- Plot (画图)



Matlab Syntax / Function

1）System defined functions

Use before help, and test on command line

```
>> help sum
 sum Sum of elements.
    S = sum(X) is the sum of the elements of the vector X. If X is a matrix,
    S is a row vector with the sum over each column. For N-D arrays,
    sum(X) operates along the first non-singleton dimension.

    S = sum(X,'all') sums all elements of X.

    S = sum(X,DIM) sums along the dimension DIM.
```

```
>> sum(A, 2)

ans =

     6
    15
    24
```

```
>> sum(A, 1)

ans =

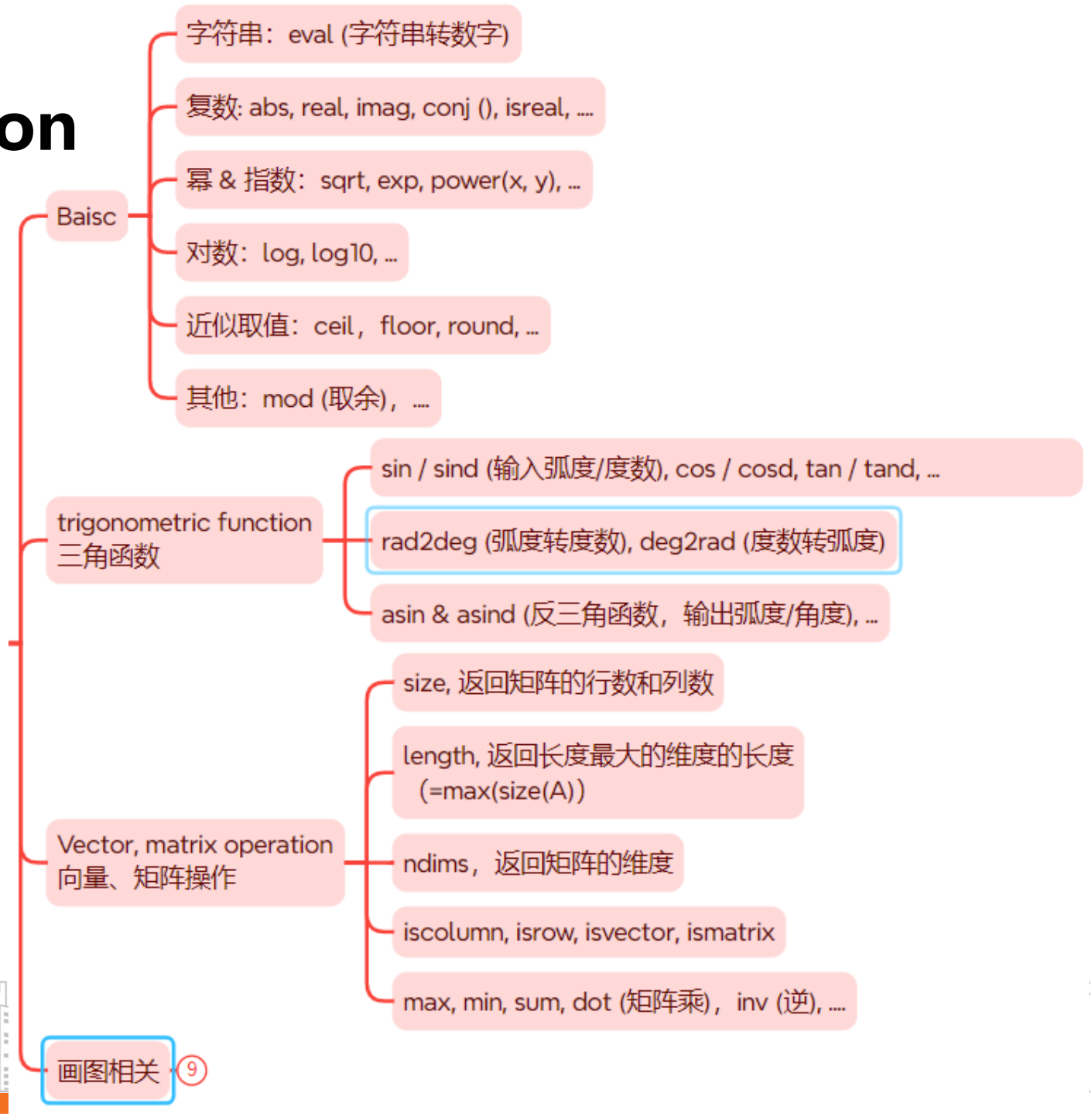
    12    15    18
```

```
>> sum(A, 'all')

ans =

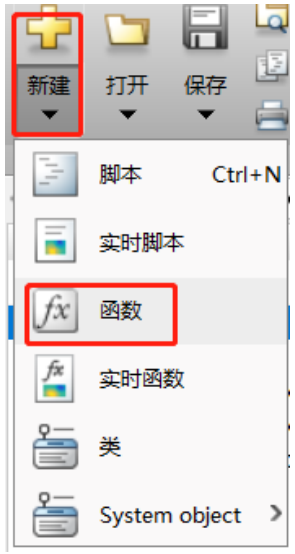
    45
```





Matlab Syntax / Function

2) Self-defined function



M script file vs M function file

- Script file: a bunch of commands, the 1st line is not a function.
- Function file: the 1st line is function, with input parameters and return values (as right). Required that function name and file name must be same.

E.g. Like myfun. m, enter myfun (x) and press enter at the command line to execute the function and return the result

%: Comment line, 注释行

lab1_my_matrix1.m

```
1  function outMatrix = lab1_my_matrix1(rowNum, colNum)
2  %  LAB1_MY_MATRIX1 生成 1个 colNum x rowNum 矩阵
3      for i = 1:rowNum
4          for j = 1:colNum
5              outMatrix(i, j) = colNum * (i -1) + j;
6          end
7      end
8  end
```

Return result declaration & Assignment

Function name And parameters list

Grammer: function ... end

Grammer: for ... end => control statements

Name Rules

- Consistent with variable naming conventions, the combination of letters, numbers, and underline (_), and starts with a letter
- DON'T be same with system defined functions

Matlab Syntax / Function

SUSTech Southern University of Science and Technology

2） Self-defined function

命令行窗口

```
>> lab1_my_matrix1(3, 3)
在当前文件夹或 MATLAB 路径中未找到 'lab1_my_matrix1'，但它位于:
 D:\work\sme206_labs\lab1_basic

更改 MATLAB 当前文件夹 或 将其文件夹添加到 MATLAB 路径。
```

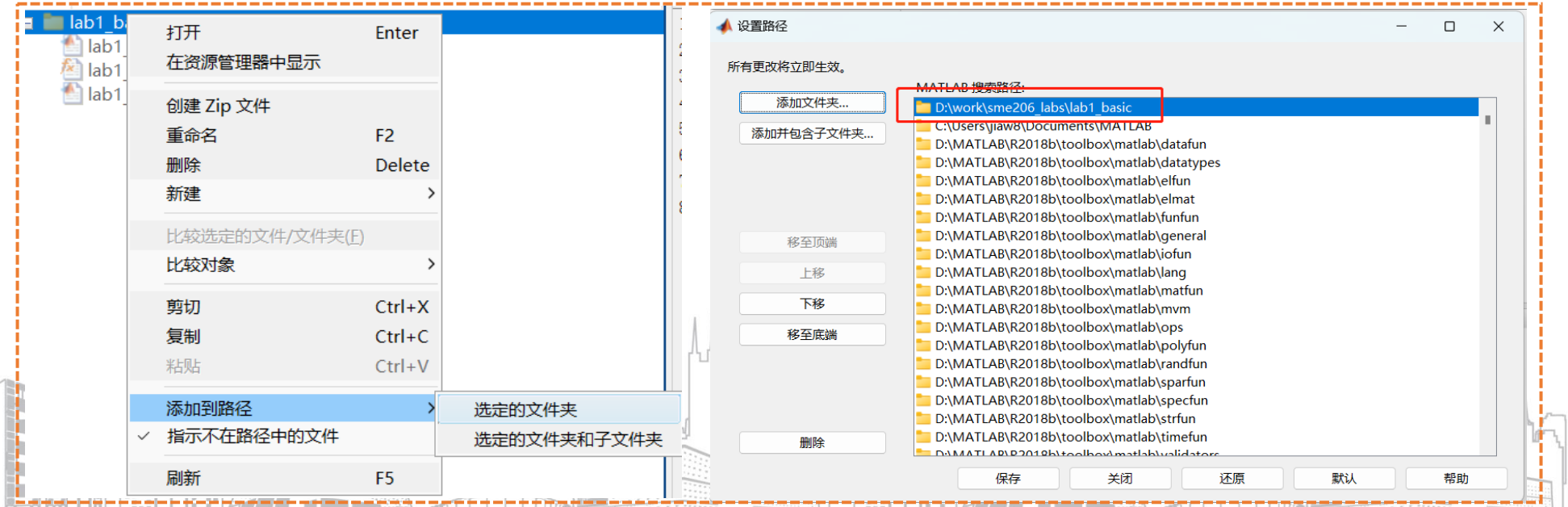
注意：
将自定义函数文件所在目录，添加到 Matlab 路径中；
或者切换当前运行目录路径。

```
>> cd

D:\work\sme206_labs
```

```
>> cd lab1_basic\
>> cd

D:\work\sme206_labs\lab1_basic
```




```
>> lab1_my_matrix1(3, 3)

ans =

     1     2     3
     4     5     6
     7     8     9
```

Matlab Syntax / Function

2) Self-defined function

Extension: Return value
- Single value of matrix
- Multiple matrices

lab1_my_piecewise_func.m

```
1   function y = lab1_my_piecewise_funct(x)
2   % lab1_my_piecewise_funct 分段函数
3   % relu,
4   %   when x <= 0, y = 0
5   %   when x > 0, y = 1
6       if x <= 0
7           y = 0;
8       else
9           y = x;
10      end
11  end
```

```
>> lab1_my_piecewise_func(10)

y =

    10
```

```
>> lab1_my_piecewise_func(-10)

y =

     0
```

lab1_my_matrix1.m | lab1_my_matrix2.m

```
1   function [A, B] = lab1_my_matrix1(rowNum, colNum)
2   %   LAB1_MY_MATRIX2 生成 2个 colNum x rowNum 矩阵
3       for i = 1:rowNum
4           for j = 1:colNum
5               A(i, j) = colNum * (i -1) + j;
6               B(i, j) = i * j;
7           end
8       end
9   end
```

```
>> [A, B] =lab1_my_matrix2(3, 4)

A =

     1     2     3     4
     5     6     7     8
     9    10    11    12


B =

     1     2     3     4
     2     4     6     8
     3     6     9    12
```

Matlab Syntax / Outline

SUSTech Southern University of Science and Technology

- Data & Variables (数据&变量)
- Operator (运算符)
- Function (函数)
- **Flow Control (流程控制)**
- Plot (画图)



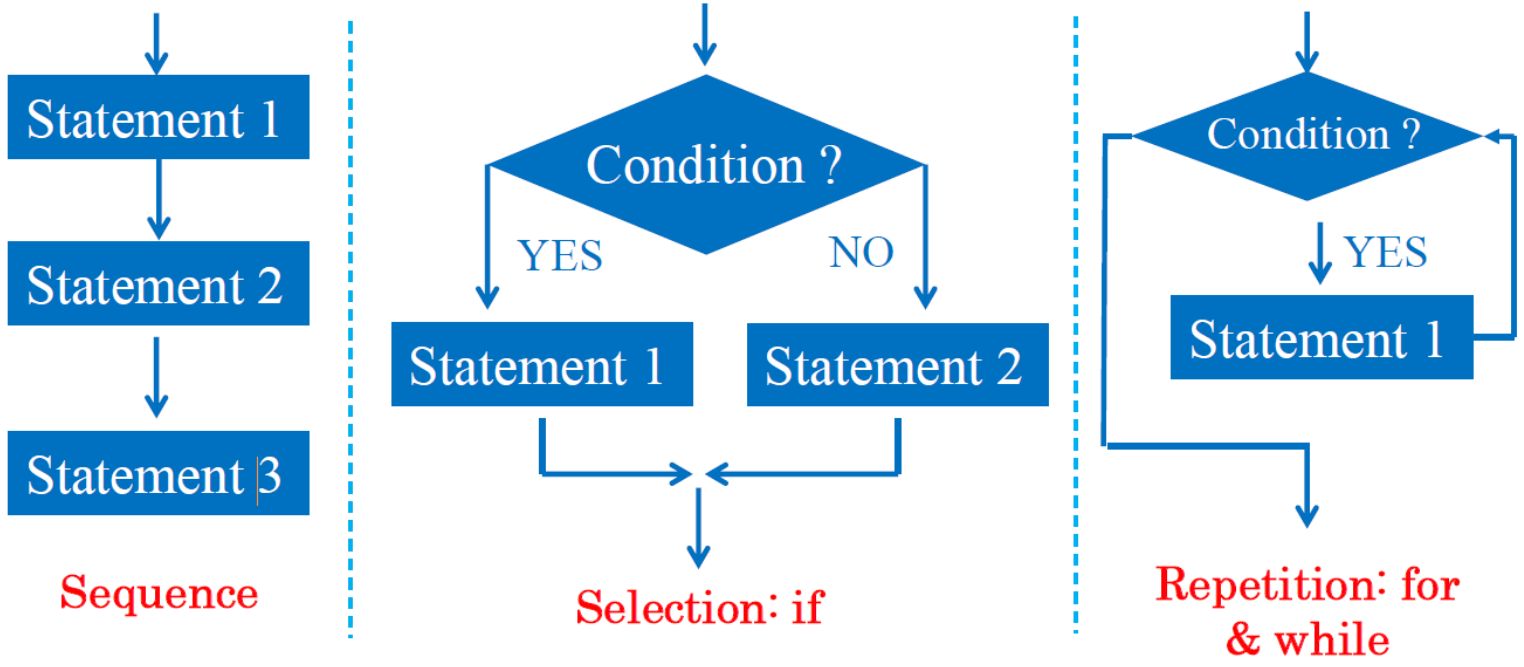
Matlab Syntax / Flow Control

SUSTech Southern University of Science and Technology

3 basic blocks

- Sequence (顺序执行)
- Condition (条件)
- Loop (循环)

=> Almost all programming languages are similar

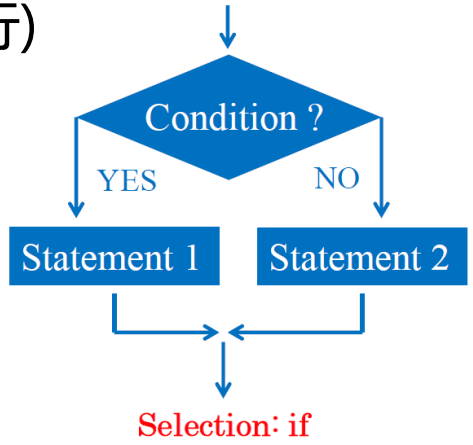


Matlab Syntax / Flow Control

SUSTech Southern University of Science and Technology

3 basic blocks

- Sequence (顺序执行)
- **Condition (条件)**
- Loop (循环)



lab1_my_piecewise_func.m

```
function y = lab1_my_piecewise_funct(x)
% lab1_my_piecewise_funct 分段函数
% relu,
%   when x <= 0, y = 0
%   when x > 0, y = 1
    if x <= 0
        y = 0;
    else
        y = x;
    end
end
```

if...elseif...else...end

lab1_my_str2val.m

```
function val = lab1_my_str2val(str)
% lab1_my_str2val, str 转为 int
    switch str
        case 'A'
            val = 1
        case 'B'
            val = 2
        case {'C', 'D'}
            val = 3
        otherwise
            val = -1
    end
end
```

switch...case...otherwise...end

⇒ case 后不接冒号 :

⇒ 多个值使用花括号 {}

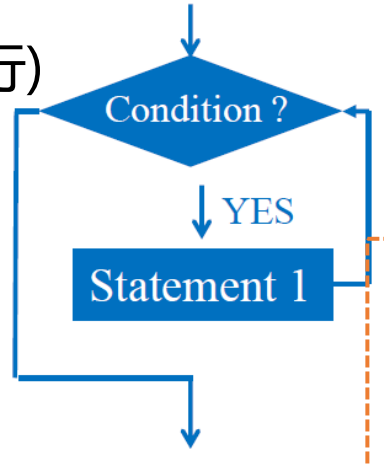
Matlab Syntax / Flow Control

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \frac{1}{11} \cdots$$

SUSTech Southern University of Science and Technology

3 basic blocks

- Sequence (顺序执行)
- Condition (条件)
- **Loop (循环)**




```
while {loop_condition}
   ...
end
```

lab1_my_pi.m

```
function result = lab1_my_pi(eps)
%lab1_my_pi π的莱布尼兹公式
result = 4.0;
last = 0.0;
diff = result - last;
n = 1;
while (diff > eps && n <= 100000)
    last = result;
    if mod(n, 2) == 1
        result = result - 4.0 / (2*n +1.0);
    else
        result = result + 4.0 / (2*n +1.0);
    end
    n = n +1;
    diff = abs(result - last);
end
end
```

lab1_my_pi.m | lab1_my_matrix1.m

```
function outMatrix = lab1_my_matrix1(rowNum, colNum)
%  LAB1_MY_MATRIX1 生成 1个 colNum x rowNum 矩阵
    for i = 1:rowNum
        for j = 1:colNum
            outMatrix(i, j) = colNum * (i -1) + j;
        end
    end
end
```

```
for i = {index_array}
   ...
end
```

```
>> lab1_my_pi(1e-6)

ans =

    3.1416
```

Matlab Syntax / Flow Control

SUSTech Southern University of Science and Technology

3 basic blocks

- Sequence (顺序执行)
- Condition (条件)
- **Loop (循环)**

Tips: Avoid looping statements as possible

⇒ Matlab is good at matrix computation

```
c=[a1+b1 a1+b2 a1+b3;
   a2+b1 a2+b2 a2+b3;
   a3+b1 a3+b2 a3+b3]
```

lab1_loop_optimize.m

```
1  A_arr = 1:100;
2  B_arr = 1:200;
3  rownum = length(A_arr);
4  colnum = length(B_arr);
5
6  disp("test add_by_loop");
7  tic;
8  C_mat = zeros(rownum, colnum);
9  for i = 1:rownum
10     for j = 1:colnum
11        C_mat(i, j) = A_arr(i) + B_arr(j);
12     end
13 end
14 toc;
15
16 disp("test add_by_matrix");
17 C_mat = ones(rownum, colnum);
18 tic;
19 C_mat = diag(A_arr) * C_mat  + C_mat * diag(B_arr);
20 toc;
```

```
>> lab1_loop_optimize
test add_by_loop
时间已过 0.001029 秒。
test add_by_matrix
时间已过 0.000573 秒。
```

Matlab Syntax / Outline

SUSTech Southern University of Science and Technology

- Data & Variables (数据&变量)
- Operator (运算符)
- Function (函数)
- Flow Control (流程控制)
- **Plot (画图)**



Matlab Syntax / Plot

SUSTech Southern University of Science and Technology

Overview

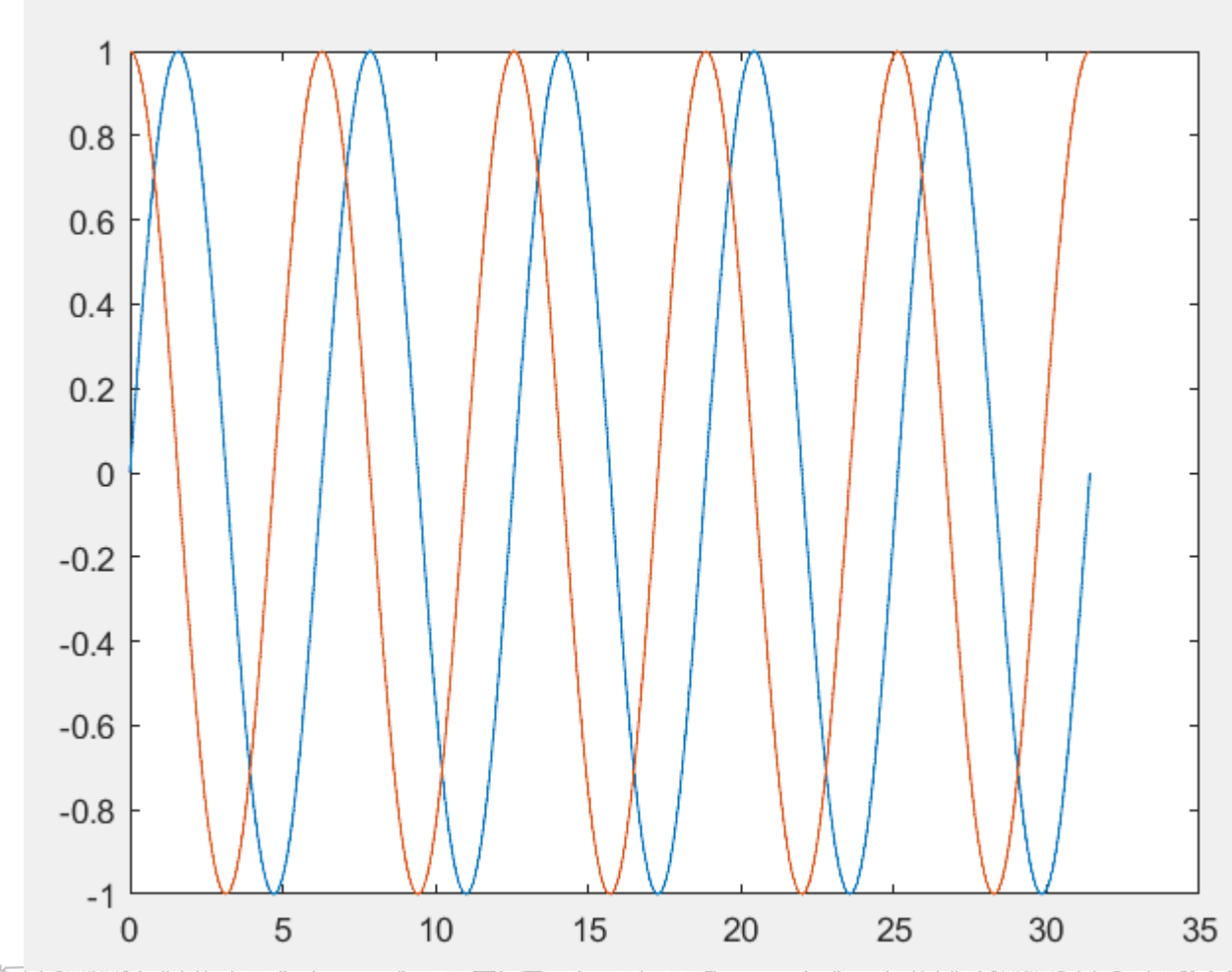
- 2D line graph (二维线图)：plot(x, y), plot(x1, y1, x2, y2, ...)
- Discrete data graph (离散数据图)：stem(火柴杆图), stairs (阶梯图), scatter (散点图), ...
- 打开 / 关闭网格线：grid on / grid off
- 横轴、纵轴标签：xlabel, ylabel
- 横轴、纵轴范围：xlim, ylim
- 添加标题：title
- 添加实例说明：legend("线条1", "线条2", ...)
- 不刷新画布 (1张画布多张图) / 刷新 .. : hold on / hold off
- 子图划分： subplot

Matlab Syntax / Plot

SUSTech Southern University of Science and Technology

Showcase: plot

```
2 -     x = linspace(0, 10*pi, 10000);
3 -     y1 = sin(x);
4 -     y2 = cos(x);
5 -     plot(x, y1, x, y2);
```



Matlab Syntax / Plot

Showcase: plot

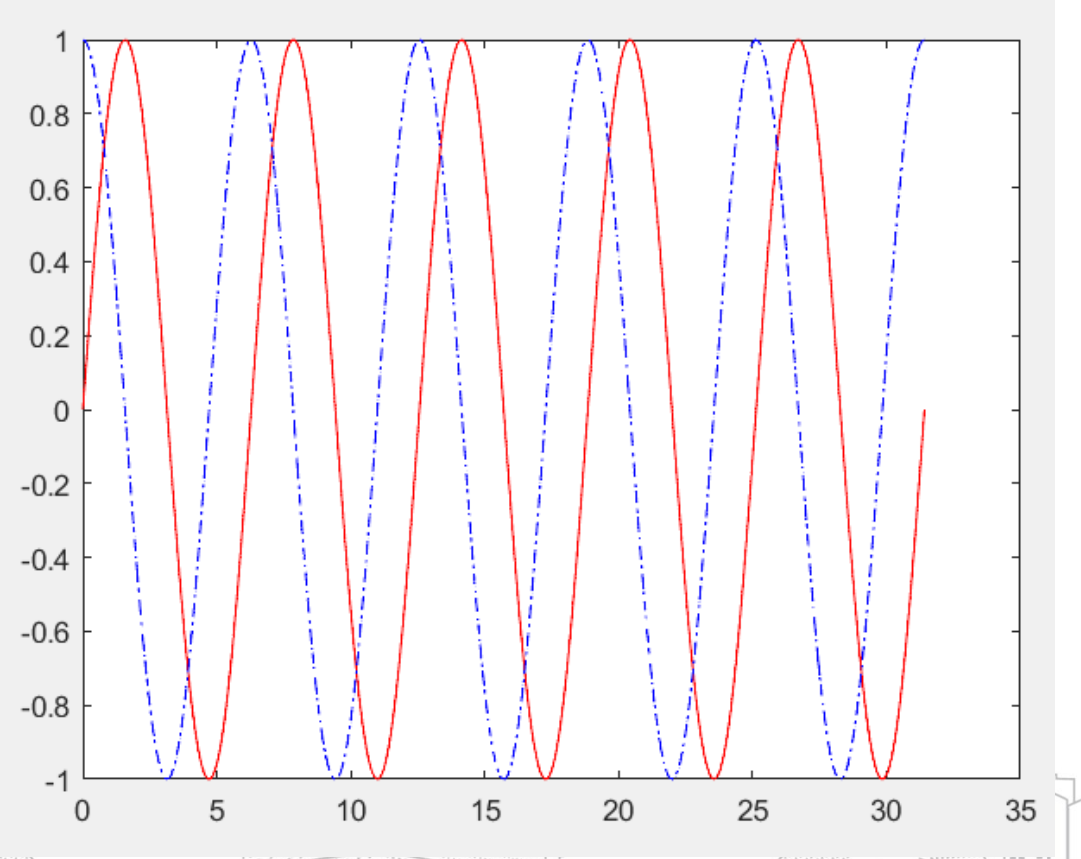
⇒ Set the line type (线型), color, and point shape (数据点形)

⇒ help plot

b	blue	.	point	-	solid
g	green	o	circle	:	dotted
r	red	x	x-mark	-.	dashdot
c	cyan	+	plus	--	dashed
m	magenta	*	star	(none)	no line
y	yellow	s	square		
k	black	d	diamond		
w	white	v	triangle (down)		
		^	triangle (up)		
		<	triangle (left)		
		>	triangle (right)		
		p	pentagram		
		h	hexagram		

For example, plot(X,Y,'c+:') plots a cyan dotted line with a plus at each data point; plot(X,Y,'bd') plots blue diamond at each data point but does not draw any line.

```
2 -     x = linspace(0, 10*pi, 10000);
3 -     y1 = sin(x);
4 -     y2 = cos(x);
5 -     plot(x, y1, '-r', x, y2, '-.b');
```



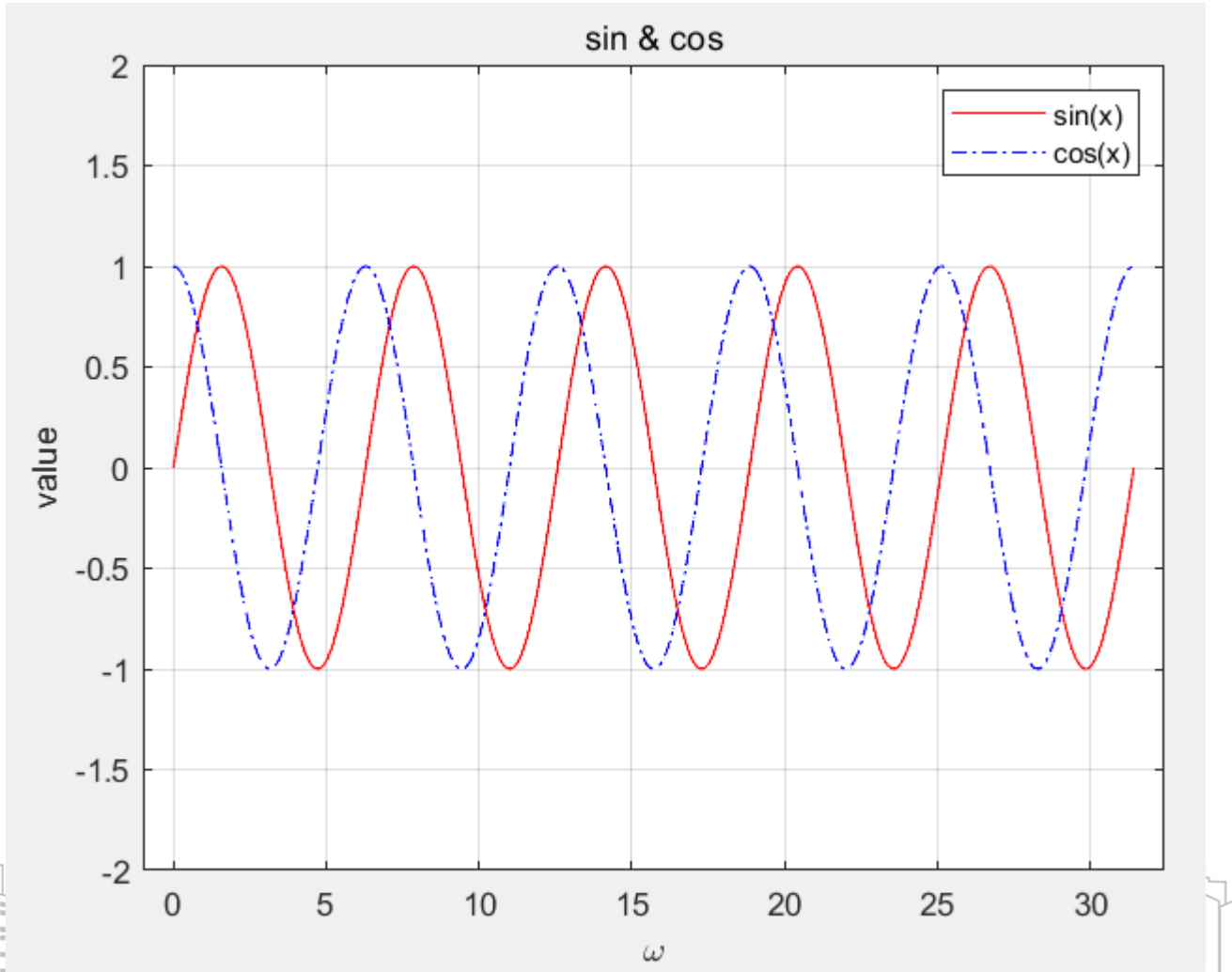
Matlab Syntax / Plot

SUSTech Southern University of Science and Technology

Showcase: plot

⇒ Set coordinate label & range, grid on/off
⇒ Set image title, and line legend (图例)

```
2 -     x = linspace(0, 10*pi, 10000);
3 -     y1 = sin(x);
4 -     y2 = cos(x);
5       % plot(x, y1, '-r', x, y2, '-.b');
6 -     plot(x, y1, '-r');
7 -     hold on;
8 -     plot(x, y2, '-.b');
9 -     xlim([-1, 10*pi + 1.0]);
10 -    ylim([-2, 2]);
11 -    xlabel("\omega");
12 -    ylabel("value");
13 -    legend('sin(x)', 'cos(x)');
14 -    title("sin & cos");
15 -    grid on;
```



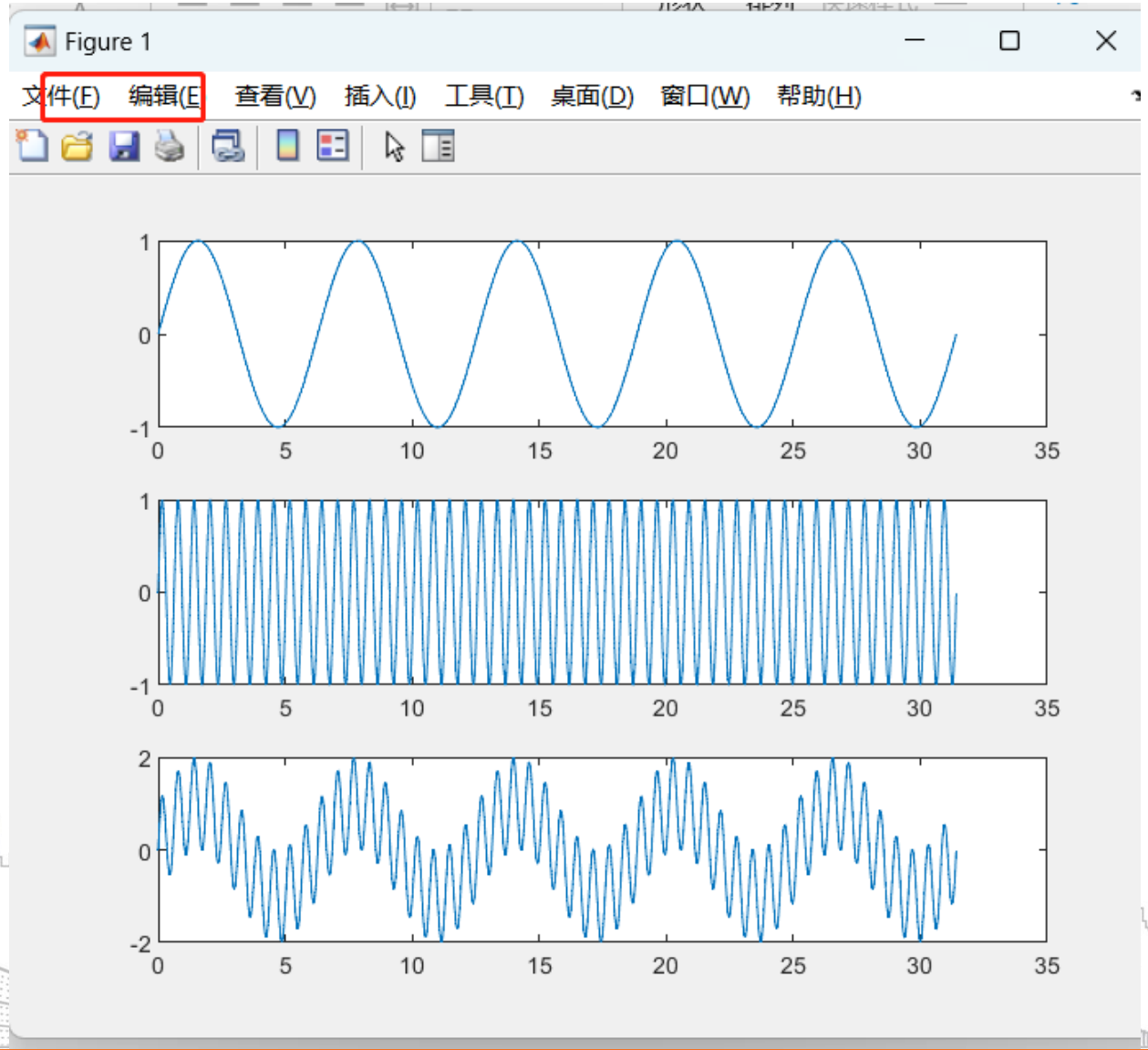
Matlab Syntax / Plot

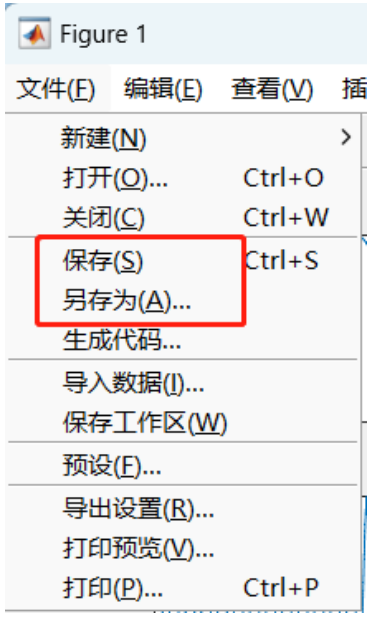
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Showcase: subplot

lab1_plot_sin_v1.m

```
1
2 -     x = linspace(0, 10*pi, 10000);
3 -     y1 = sin(x);
4 -     y2 = sin(10 * x);
5 -     y3 = y1 +y2;
6 -     subplot(3, 1, 1);
7 -     plot(x, y1);
8 -     subplot(3, 1, 2);
9 -     plot(x, y2);
10 -    subplot(3, 1, 3);
11 -    plot(x, y3);
```



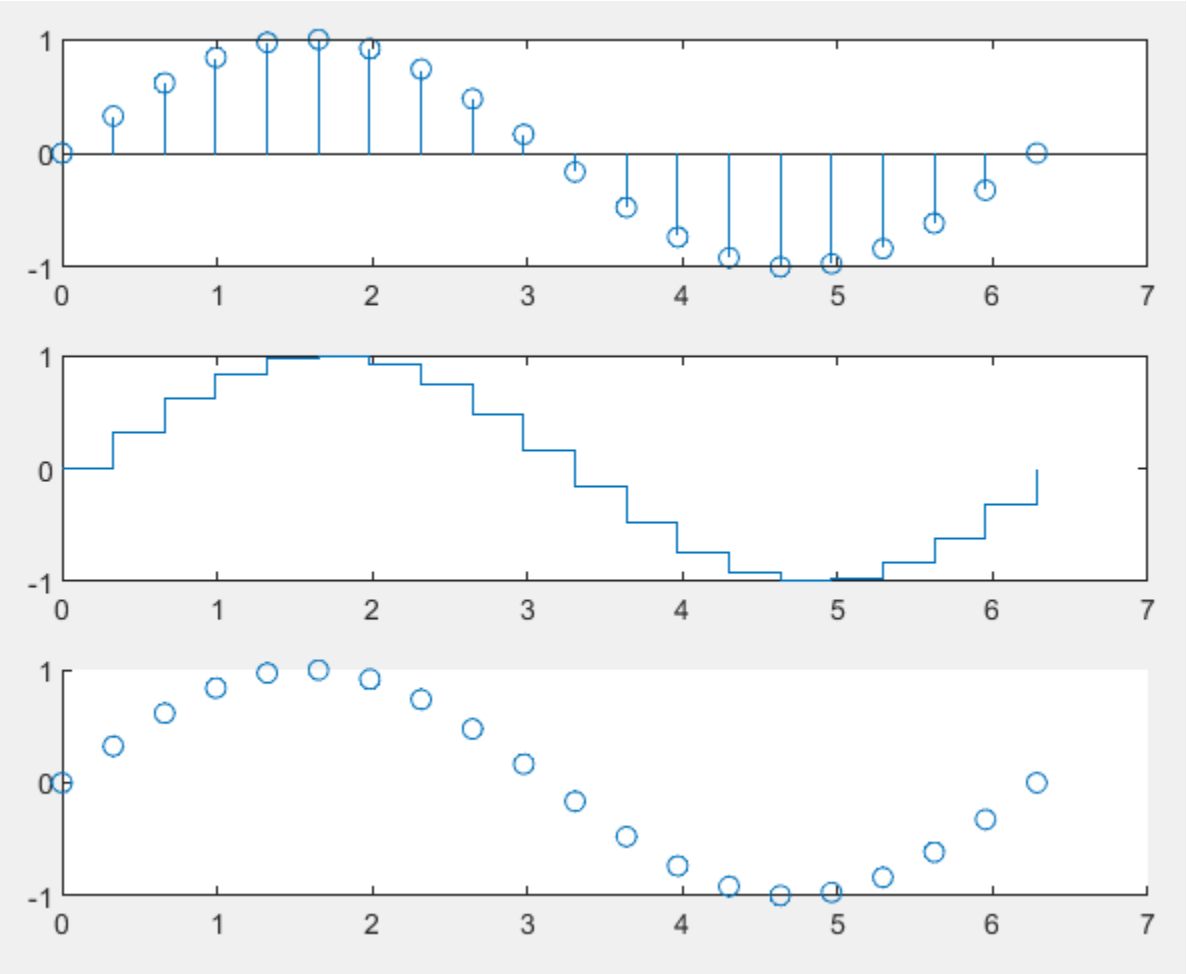


Matlab Syntax / Plot

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Showcase: stem, stairs, scatter

```
2 -     x = linspace(0, 2*pi, 20);
3 -     y1 = sin(x);
4 -     subplot(3, 1, 1);
5 -     stem(x, y1);
6 -     subplot(3, 1, 2);
7 -     stairs(x, y1);
8 -     subplot(3, 1, 3);
9 -     scatter(x, y1);
```



Lab1 / Task 1

1.1 Tutorial: Basic Matlab functions for Representing Signals

⇒ 离散信号表示

$$x[n] = \begin{cases} 2n, & -3 \le n \le 3, \\ 0, & \text{otherwise}, \end{cases}$$

⇒ 复数表示

```
>> n = [0:32];
>> x = exp(j*(pi/8)*n);
```

⇒ 运算符：加、减、乘、除……

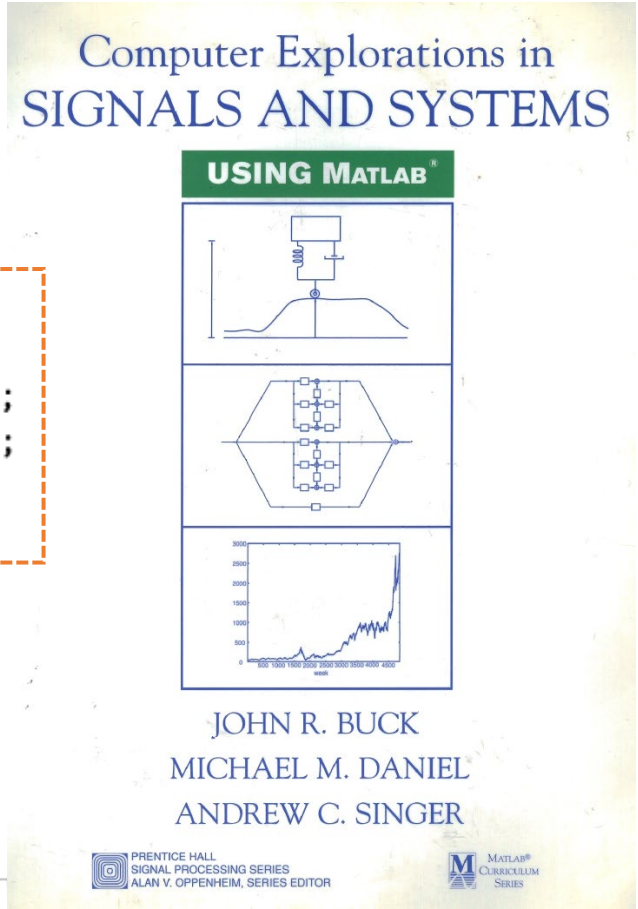
```
>> y1 = x1+x2;
>> y2 = x1-x2;
>> y3 = x1.*x2;
>> y4 = x1./x2;
>> y5 = 2*x1;
>> y6 = x1.^3;
```

⇒ 画图：stem 使用

```
% prob1.m
n = [0:16];
x1 = cos(pi*n/4);
y1 = mean(x1);
stem(n,x1)
title('x1 = cos(pi*n/4)')
xlabel('n (samples)')
ylabel('x1[n]')
```

⇒ 函数：foo 例子

```
function [y,z] = foo(x)
% [y,z] = foo(x) accepts a numerical argument x and
% returns two arguments y and z, where y is 2*x
% and z is (5/9)*(x-32)
y = 2*x;
z = (5/9)*(x-32);
```



Outline

SUSTech Southern University of Science and Technology

- Lab Intro
- Matlab Intro
- Matlab Syntax
- **Signals Basic**



Signals Basic / Topics

SUSTech Southern University of Science and Technology

- 离散时间正弦信号，Discrete-Time Sinusoidal Signals
- 连续时间复指数信号，Continuous-Time Complex Exponential Signals
- 离线时间信号的时间变换，Transformation of the Time Index for Discrete-Time Signals
- 连续时间信号的时间变换，Transformations of the Time index for Continuous-Time Signals
- 离散时间系统的性质，Properties of Discrete-Time Systems



Signals Basic / Discrete-Time Sinusoidal Signals (a)

From Lab Book, 1.2)

Basic Problems

(a). Consider the discrete-time signal

$$x_M[n] = \sin\left(\frac{2\pi M n}{N}\right),$$

and assume $N = 12$. For $M = 4$, 5, 7, and 10, plot $x_M[n]$ on the interval $0 \leq n \leq 2N - 1$. Use `stem` to create your plots, and be sure to appropriately label your axes. What is the fundamental period of each signal? In general, how can the fundamental period be determined from arbitrary integer values of M and N? Be sure to consider the case in which $M > N$.

Signals Basic / Discrete-Time Sinusoidal Signals (a)

From Lab Book, 1.2)

Basic Problems

(a). Consider the discrete-time signal

$$x_M[n] = \sin\left(\frac{2\pi M n}{N}\right),$$

and assume $N = 12$. For $M = 4$, 5, 7, and 10, plot $x_M[n]$ on the interval $0 \leq n \leq 2N - 1$. Use `stem` to create your plots, and be sure to appropriately label your axes. What is the fundamental period of each signal? In general, how can the fundamental period be determined from arbitrary integer values of M and N? Be sure to consider the case in which $M > N$.

```
N = 12;
n = 0: 2*N-1;

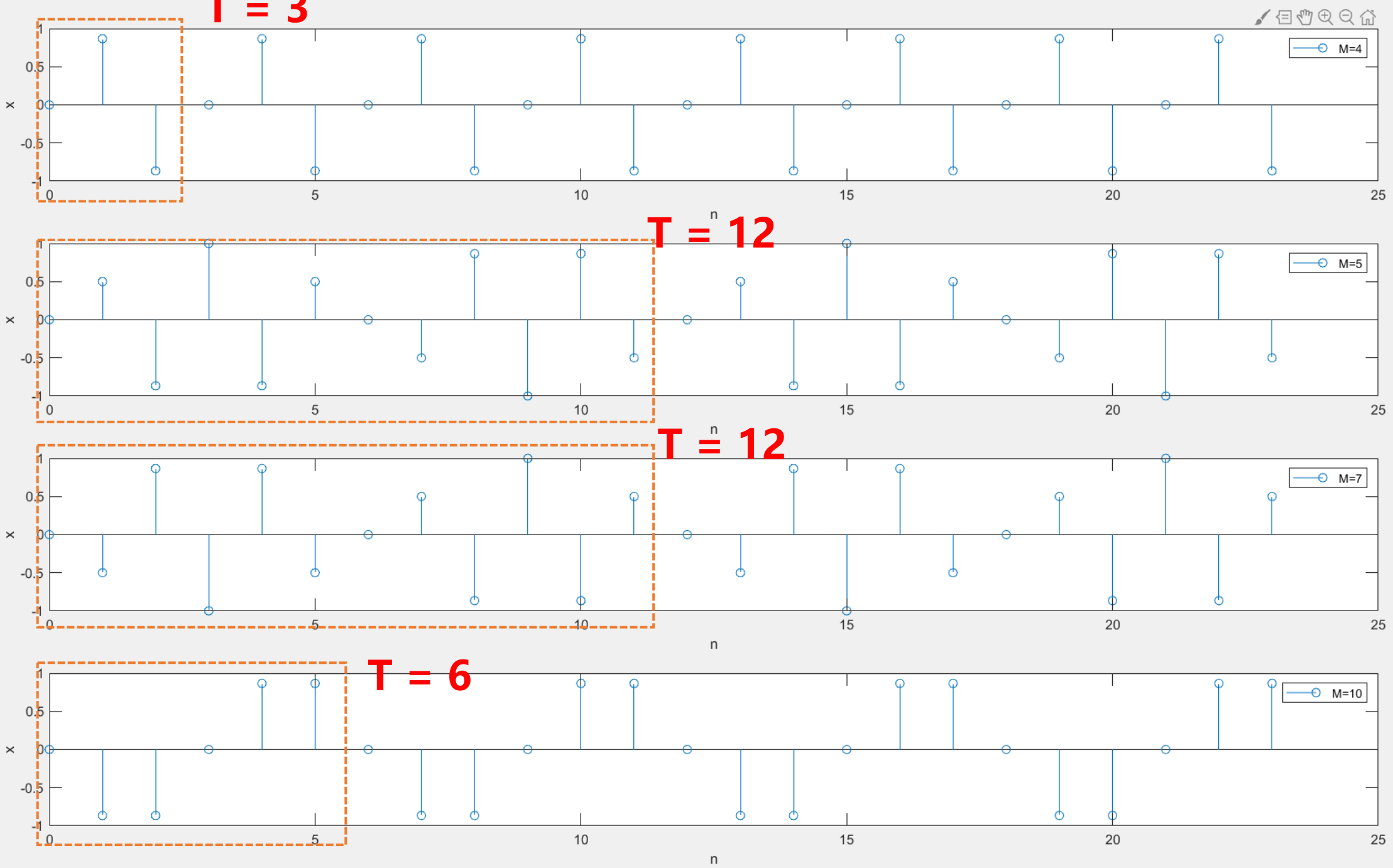
x4 = sin(2*pi *4 * n / N);
x5 = sin(2*pi *5 * n / N);
x7 = sin(2*pi *7 * n / N);
x10 = sin(2*pi *10 * n / N);

% problem a
subplot(4, 1, 1);
stem(n, x4);
xlabel("n");
ylabel("x");
legend('M=4');

subplot(4, 1, 2);
stem(n, x5);
xlabel("n");
ylabel("x");
legend('M=5');

subplot(4, 1, 3);
stem(n, x7);
xlabel("n");
ylabel("x");
legend('M=7');

subplot(4, 1, 4);
stem(n, x10);
xlabel("n");
ylabel("x");
legend('M=10');
```


T = 3
M=4
T = 12
M=5
T = 12
M=7
T = 6
M=10
x
n
1
0.5
0
-0.5
-1
0
5
10
15
20
25

Signals Basic / Discrete-Time Sinusoidal Signals (a)

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Q, what is the fundamental period of each signal?

=> The smallest integer of k * N/M, where k = 1, 2, ...

Proof:

$$\sin\frac{2\pi M n}{N} = \sin\frac{2\pi M (n+T_0)}{N} = \sin\left(\frac{2\pi M n}{N} + \frac{2\pi M T_0}{N}\right)$$

⇨ T_0 需满足: $\frac{2\pi M T_0}{N} = 2\pi k,\ \ k = 1, 2, 3, \ldots$

⇨ 于是: $T_0 = k\frac{N}{M}$

Signals Basic / Discrete-Time Sinusoidal Signals (c)

SUSTech Southern University of Science and Technology

From Lab Book, 1.2)

(c). Now consider the following three signals

$$x_1[n] = \cos\left(\frac{2\pi n}{N}\right) + 2\cos\left(\frac{3\pi n}{N}\right),$$

$$x_2[n] = 2\cos\left(\frac{2n}{N}\right) + \cos\left(\frac{3n}{N}\right),$$

$$x_3[n] = \cos\left(\frac{2\pi n}{N}\right) + 3\sin\left(\frac{5\pi n}{2N}\right).$$

Assume $N = 6$ for each signal. Determine whether or not each signal is periodic. If a signal is periodic, plot the signal for two periods, starting at $n = 0$. If the signal is not periodic, plot the signal for $0 \leq n \leq 4N$ and explain why it is not periodic. Remember to use `stem` and to appropriately label your axes.

Signals Basic / Discrete-Time Sinusoidal Signals (c)

SUSTech Southern University of Science and Technology

From Lab Book, 1.2)

(c). Now consider the following three signals

$$x_1[n] = \cos\left(\frac{2\pi n}{N}\right) + 2\cos\left(\frac{3\pi n}{N}\right),$$

$$x_2[n] = 2\cos\left(\frac{2n}{N}\right) + \cos\left(\frac{3n}{N}\right),$$

$$x_3[n] = \cos\left(\frac{2\pi n}{N}\right) + 3\sin\left(\frac{5\pi n}{2N}\right).$$

Assume $N = 6$ for each signal. Determine whether or not each signal is periodic. If a signal is periodic, plot the signal for two periods, starting at $n = 0$. If the signal is not periodic, plot the signal for $0 \leq n \leq 4N$ and explain why it is not periodic. Remember to use `stem` and to appropriately label your axes.

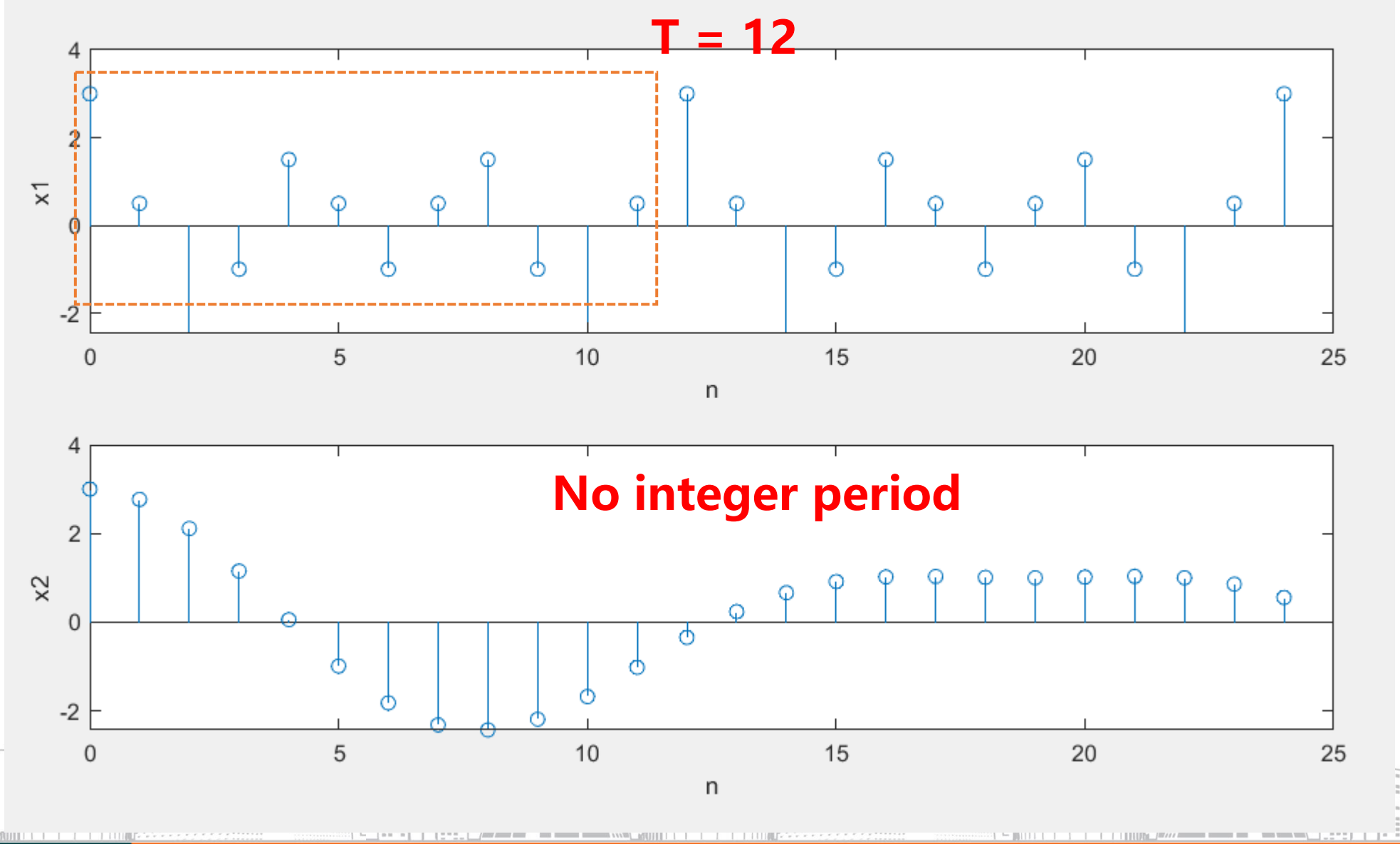
```
N = 6;
n = 0: 4*N;

x1 = cos(2*pi * n / N) + 2 * cos(3*pi * n / N);
x2 = 2 * cos(2* n / N) + cos(3* n / N);

% problem c
subplot(3, 1, 1);
stem(n, x1);
xlabel("n");
ylabel("x1");

subplot(3, 1, 2);
stem(n, x2);
xlabel("n");
ylabel("x2");
```

Signals Basic / Discrete-Time Sinusoidal Signals (c)



Signals Basic / Discrete-Time Sinusoidal Signals (c)

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Q, Determine whether or not each signal is periodic?

⇒ For the linear combination of 2 signals, take the least common multiple of the 2 signal periods

Proof:

$$\sin\frac{2\pi n}{6}+\cos\frac{3\pi n}{6}=\sin\left(\frac{2\pi n}{6}+\frac{2\pi T_0}{6}\right)+\cos\left(\frac{3\pi n}{6}+\frac{3\pi T_0}{6}\right)$$

⇨ T_0 需满足以下 2 个条件:

$$\frac{2\pi T_0}{6}=2\pi k_1,\quad k_1=1,2,3,\ldots$$

$$\frac{3\pi T_0}{6}=2\pi k_2,\quad k_2=1,2,3,\ldots$$

⇨ 于是:

$$T_0=6k_1,\quad k_1=1,2,3,\ldots$$

$$T_0=4k_2,\quad k_2=1,2,3,\ldots$$

当 $k_1=1,\ k_2=3$ 时, $T_0=12$ 为最小周期 (基波周期)

$\sin\frac{2n}{6}+\cos\frac{3n}{6}$, 类似

⇨ T_0 需满足以下 2 个条件:

$$T_0=6\pi k_1,\quad k_1=1,2,3,\ldots$$

$$T_0=4\pi k_2,\quad k_2=1,2,3,\ldots$$

显然, 不存在整数 k_1、k_2, 使得 T_0 为整数

Signals Basic / Discrete-Time Sinusoidal Signals (d)

SUSTech Southern University of Science and Technology

From Lab Book, 1.2)

Intermediate Problems

(d). Plot each of the following signals on the interval $0 \leq n \leq 31$:

$$x_1[n] = \sin\left(\frac{\pi n}{4}\right) \cos\left(\frac{\pi n}{4}\right),$$

$$x_2[n] = \cos^2\left(\frac{\pi n}{4}\right),$$

$$x_3[n] = \sin\left(\frac{\pi n}{4}\right) \cos\left(\frac{\pi n}{8}\right).$$

What is the fundamental period of each signal? For each of these three signals, how could you have determined the fundamental period without relying upon MATLAB?

(e). Consider the signals you plotted in Parts (c) and (d). Is the addition of two periodic signals necessarily periodic? Is the multiplication of two periodic signals necessarily periodic? Clearly explain your answers.

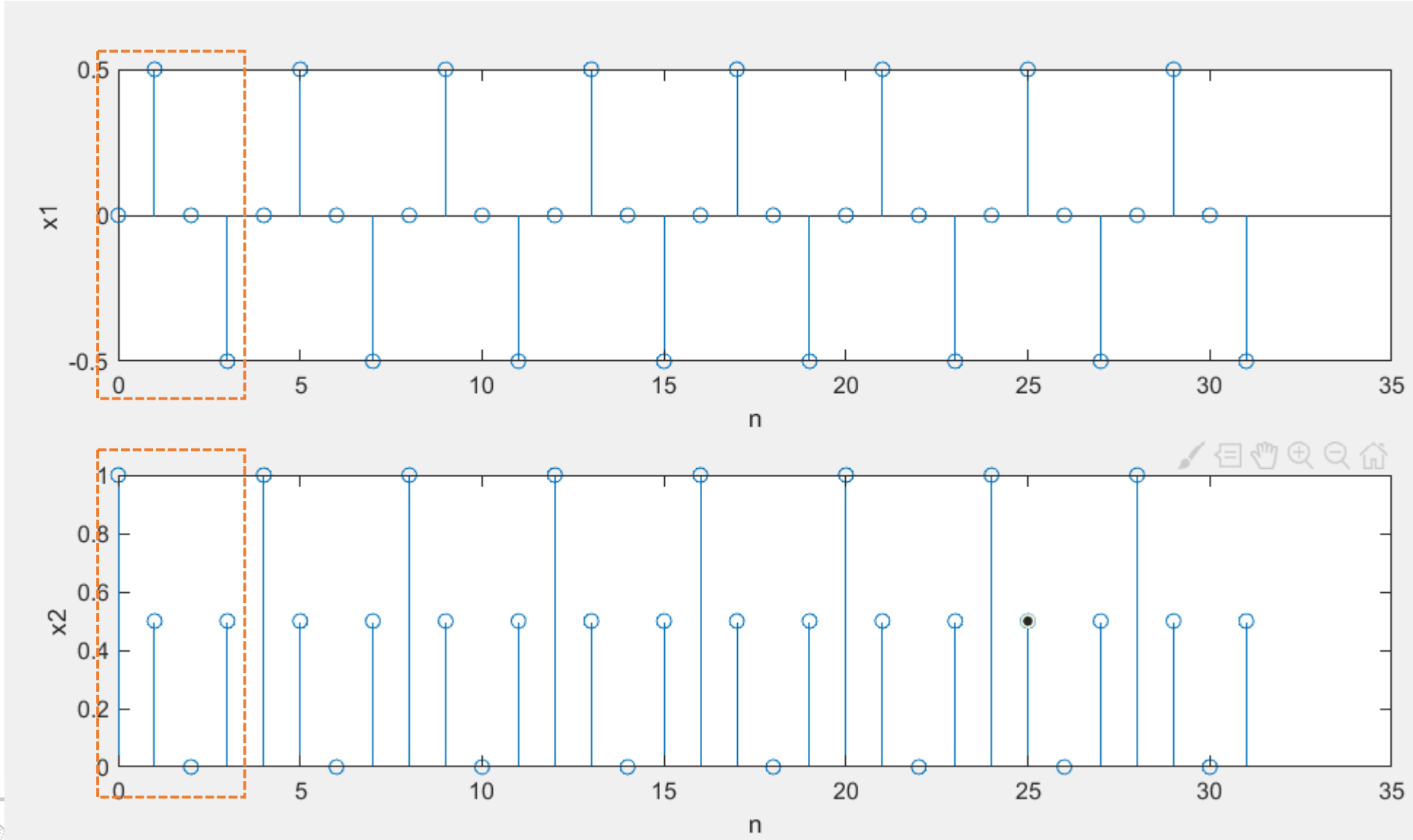
```
2 -  n = 0: 31;
3
4 -  x1 = sin(pi * n / 4) .* cos(pi * n / 4);
5 -  x2 = cos(pi * n / 4) .^ 2;
6
7    % problem d
8 -  subplot(3, 1, 1);
9 -  stem(n, x1);
10 - xlabel("n");
11 - ylabel("x1");
12
13 - subplot(3, 1, 2);
14 - stem(n, x2);
15 - xlabel("n");
16 - ylabel("x2");
```

Signals Basic / Discrete-Time Sinusoidal Signals (d)

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T = 4

T = 4



Signals Basic / Discrete-Time Sinusoidal Signals (d)

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Proof:

$$\sin\frac{\pi n}{4}\cos\frac{\pi n}{4}=\frac{1}{2}\sin(2\times\frac{\pi n}{4})=\frac{1}{2}\sin(\frac{\pi n}{2}) \Rightarrow T_0=k\frac{2\pi}{\frac{\pi}{2}}=4k$$

$$\cos^2\frac{\pi n}{4}=\frac{\cos(2\times\frac{\pi n}{4})+1}{2}=\frac{\cos(\frac{\pi n}{2})+1}{2}$$

$\cos(\alpha+\beta)=\cos\alpha\cdot\cos\beta-\sin\alpha\cdot\sin\beta$

$\cos(\alpha-\beta)=\cos\alpha\cdot\cos\beta+\sin\alpha\cdot\sin\beta$

$\sin(\alpha\pm\beta)=\sin\alpha\cdot\cos\beta\pm\cos\alpha\cdot\sin\beta$

$$\sin\alpha\cos\beta=\frac{\sin(\alpha+\beta)+\sin(\alpha-\beta)}{2}$$

$$\cos\alpha\sin\beta=\frac{\sin(\alpha+\beta)-\sin(\alpha-\beta)}{2}$$

$$\cos\alpha\cos\beta=\frac{\cos(\alpha+\beta)+\cos(\alpha-\beta)}{2}$$

$$\sin\alpha\sin\beta=-\frac{\cos(\alpha+\beta)-\cos(\alpha-\beta)}{2}$$

$$\sin 2\alpha=\sin\alpha\cos\alpha+\sin\alpha\cos\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha=2\cos^2\alpha-1=1-2\sin^2\alpha$$

Signals Basic / Topics

SUSTech Southern University of Science and Technology

- 离散时间正弦信号，Discrete-Time Sinusoidal Signals
- **连续时间复指数信号，Continuous-Time Complex Exponential Signals**
- 离线时间信号的时间变换，Transformation of the Time Index for Discrete-Time Signals
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- 离散时间系统的性质，Properties of Discrete-Time Systems



Signal Basic / Continuous-Time Complex Exponential Signals (a)

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From Lab Book, 1.6)

Basic Problems

(a). Consider the continuous-time sinusoid

$$x(t) = \sin(2\pi t/T) .$$

A symbolic expression can be created to represent $x(t)$ within MATLAB by executing

Signal Basic / Continuous-Time Complex Exponential Signals (a)

From Lab Book, 1.6)

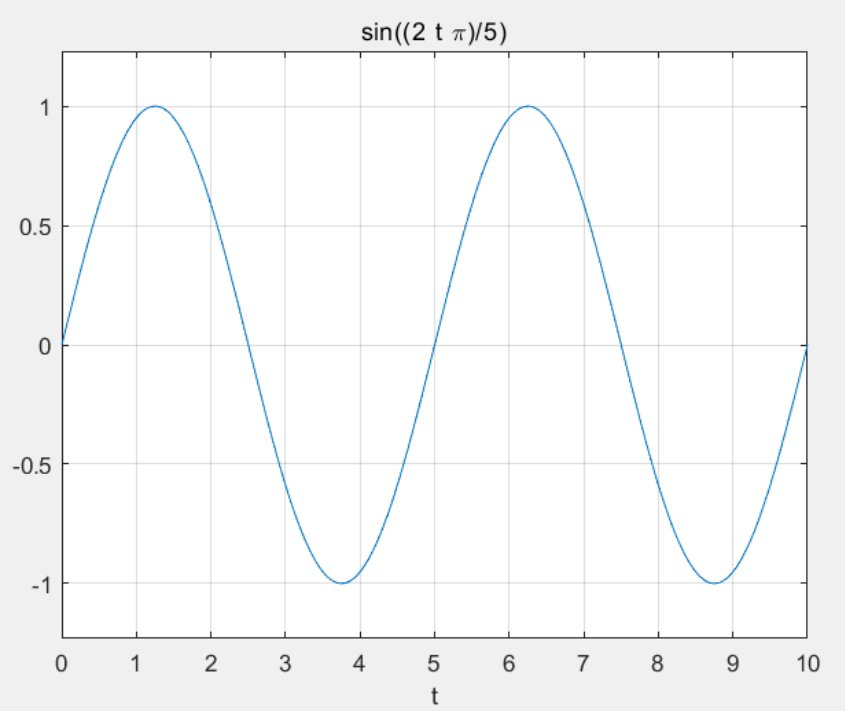
Basic Problems

(a). Consider the continuous-time sinusoid

$$x(t) = \sin(2\pi t/T)\,.$$

A symbolic expression can be created to represent $x(t)$ within MATLAB by executing

```
2 -  clear;clc;
3
4 -  syms t;
5 -  syms T;
6 -  x = sym( sin(2*pi*t/T) );
7
8 -  x5 = subs(x, T, 5);
9 -  ezplot(x5, [0, 10]);
10 - grid on;
```



Function: sym, syms, ezplot

- **syms**, to create a symbolic variable
- **sym**, which combines variables created by syms to form a new expression
- **ezplot**, plot against a **sym**

Signal Basic / Continuous-Time Complex Exponential Signals (b)

From Lab Book, 1.6)

(b). Create a symbolic expression for the signal

$$x(t) = \cos(2\pi t/T)\,\sin(2\pi t/T)\,.$$

The two sinusoids should be created separately, and then combined using `symmul`. For $T = 4$, 8, and 16, use `ezplot` to plot the signal on the interval $0 \le t \le 32$. What is the fundamental period of $x(t)$ in terms of T?

syntax update issues,
symmul(..) replace by times(..)

Functions

Arithmetic Operations

minus	Symbolic subtraction
plus	Symbolic addition
times	Symbolic array multiplication
ldivide	Symbolic array left division
rdivide	Symbolic array right division
power	Symbolic array power

https://ww2.mathworks.cn/help/symbolic/operators.html

```
4 -  syms t;
5 -  syms T;
6
7 -  s = sym( sin(2*pi*t/T) );
8 -  c = sym( cos(2*pi*t/T) );
9 -  x = times(s, c);
10
11 -  x4 = subs(x, T, 4);
12 -  x8 = subs(x, T, 8);
13 -  x16 = subs(x, T, 16);
14
15 -  subplot(3, 1, 1);
16 -  ezplot(x4, [0, 32]);
17 -  grid on;
18
19 -  subplot(3, 1, 2);
20 -  ezplot(x8, [0, 32]);
21 -  grid on;
22
23 -  subplot(3, 1, 3);
24 -  ezplot(x16, [0, 32]);
25 -  grid on;
```

Substitution
(符号替换)

Signal Basic / Continuous-Time Complex Exponential Signals (b)

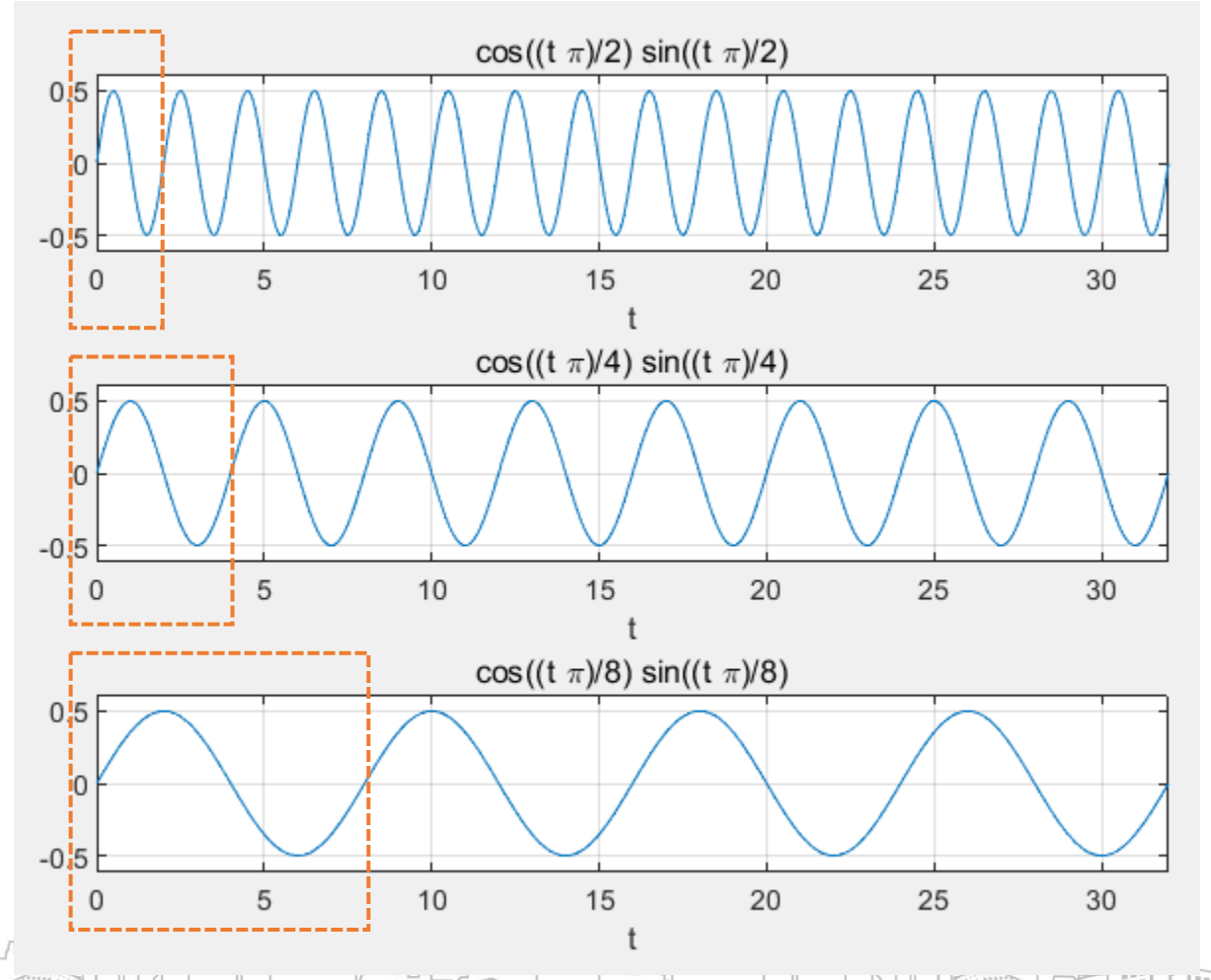
$$\sin\frac{2\pi\, t}{T}\cos\frac{2\pi\, t}{T} = \frac{1}{2}\sin\left(2\times\frac{2\pi\, t}{T}\right) = \frac{1}{2}\sin\left(\frac{4\pi\, t}{T}\right)$$

$$\Rightarrow\quad T_0 = \frac{2\pi}{\frac{4\pi}{T}} = \frac{T}{2}$$

T0 = 2

T0 = 4

T0 = 8



Signals Basic / Topics

SUSTech Southern University of Science and Technology

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Signals Basic / Transformation of the Time Index for DT Signals (a)

SUSTech Southern University of Science and Technology

From Lab Book, 1.3)

Basic Problems

(a). Define a MATLAB vector **nx** to be the time indices $-3 \leq n \leq 7$ and the MATLAB vector **x** to be the values of the signal $x[n]$ at those samples, where $x[n]$ is given by

$$x[n] = \begin{cases} 2, & n = 0, \\ 1, & n = 2, \\ -1, & n = 3, \\ 3, & n = 4, \\ 0, & \text{otherwise}. \end{cases}$$

If you have defined these vectors correctly, you should be able to plot this discrete-time sequence by typing `stem(nx,x)`. The resulting plot should match the plot shown in Figure 1.1.

lab1_dt_tranf_abc.m | lab1_dt_tranf_func.m

```
1   % lab book 1.3,  transformations of the t
2   % base function
3   function y = lab1_dt_tranf_func(n)
4       if n == 0
5           y = 2
6       elseif n == 2
7           y = 1
8       elseif n == 3
9           y = -1
10      elseif n == 4
11          y = 3
12      else
13          y = 0
14      end
15  end
```

Signals Basic / Transformation of the Time Index for DT Signals (b, c)

SUSTech Southern University of Science and Technology

From Lab Book, 1.4)

(b). For this part, you will define MATLAB vectors `y1` through `y4` to represent the following discrete-time signals

$$y_1[n] = x[n-2],$$
$$y_2[n] = x[n+1],$$
$$y_3[n] = x[-n],$$
$$y_4[n] = x[-n+1].$$

To do this, you should define `y1` through `y4` to be equal to `x`. The key is to define correctly the corresponding index vectors `ny1` through `ny4`. First, you should figure out how the index of a given sample of $x[n]$ changes when transforming to $y_i[n]$. The index vectors need not span the same set of indices as `nx`, but they should all be at least 11 samples long and include the indices of all nonzero samples of the associated signal.

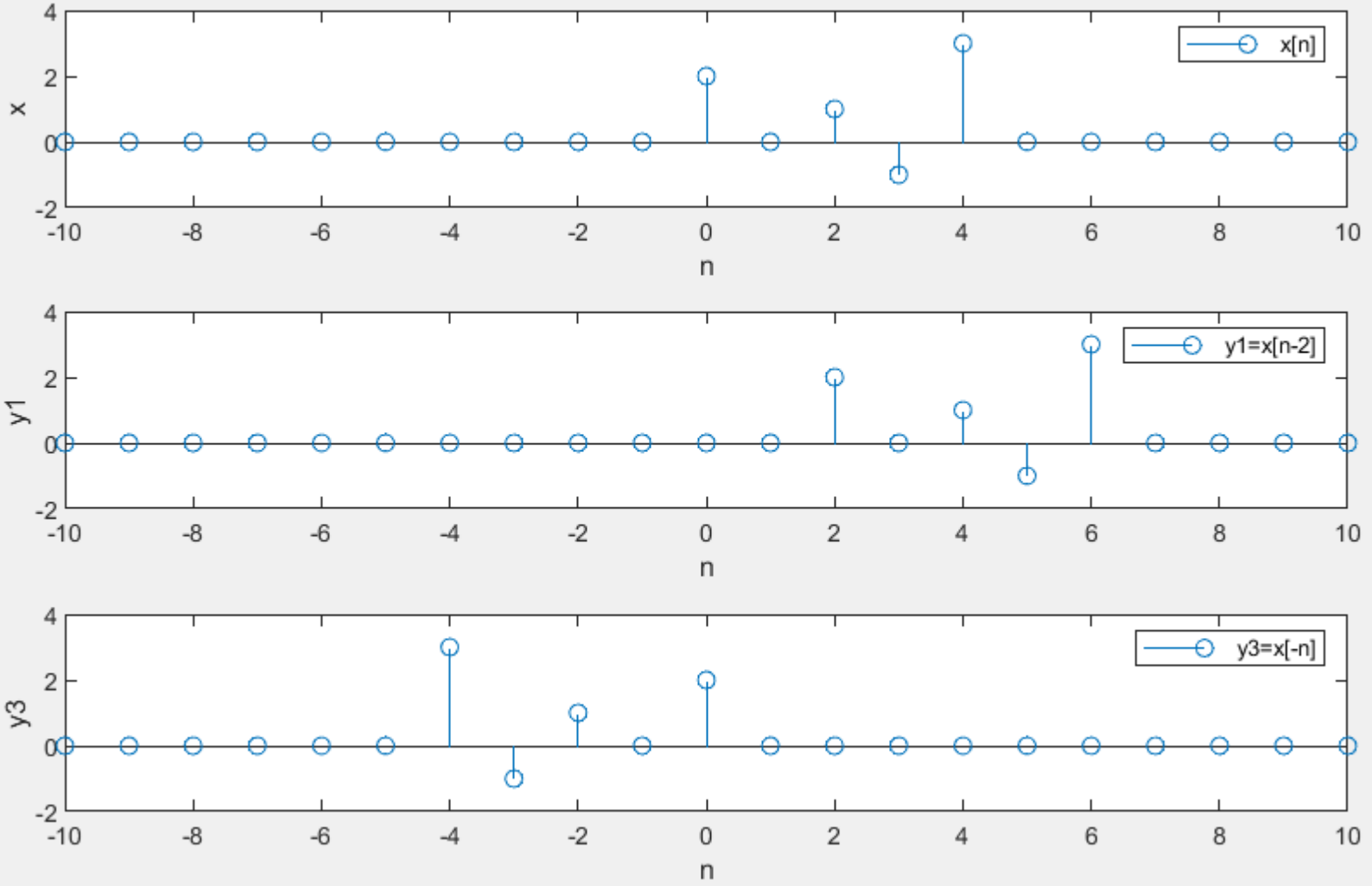
(c). Generate plots of $y_1[n]$ through $y_4[n]$ using `stem`. Based on your plots, state how each signal is related to the original $x[n]$, e.g., "delayed by 4" or "flipped and then advanced by 3."

```
4 -  n = -10:10
5    % x = lab1_dt_tranf_func(n)
6 -  x = arrayfun(@lab1_dt_tranf_func, n)
7
8 -  y1 = arrayfun(@lab1_dt_tranf_func, n-2)
9 -  y3 = arrayfun(@lab1_dt_tranf_func, -n)
10
11 - subplot(4, 1, 1);
12 - stem(n, x);
13 - xlabel("n");
14 - ylabel("x");
15 - legend('x[n]');
16
17 - subplot(4, 1, 2);
18 - stem(n, y1);
19 - xlabel("n");
20 - ylabel("y1");
21 - legend('y1=x[n-2]');
22
23 - subplot(4, 1, 3);
24 - stem(n, y3);
25 - xlabel("n");
26 - ylabel("y3");
27 - legend('y3=x[-n]');
```

Signals Basic / Transformation of the Time Index for DT Signals (b, c)

Shift 2 units to the right

Flip horizontally (centered on x=0)



Extension: Matlab function optimization for vector input

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when n = -10:10, input into following function，why the output is 0?

```
3   function y = lab1_dt_tranf_func(n)
4       if n == 0
5           y = 2
6       elseif n == 2
7           y = 1
8       elseif n == 3
9           y = -1
10      elseif n == 4
11          y = 3
12      else
13          y = 0
14      end
15  end
```

```
>> lab1_dt_tranf_func(n);

y =

     0
```

why?

If change to following optimized functions, and the output is as expected !

lab1_dt_tranf_func_opt.m

```
1   % lab book 1.3,  transformations of the time index for Di
2   % base function
3   function y = lab1_dt_tranf_func_opt(n)
4       y = 2 * (n == 0) + (n ==2) - (n == 3) + 3 * (n == 4);
5   end
```

```
>> lab1_dt_tranf_func_opt(n)

ans =

 1 至 15 列

   0   0   0   0   0   0   0   0   0   0   2   0   1  -1   3

 16 至 21 列

   0   0   0   0   0   0
```

Signals Basic / Transformation of the Time Index for DT Signals

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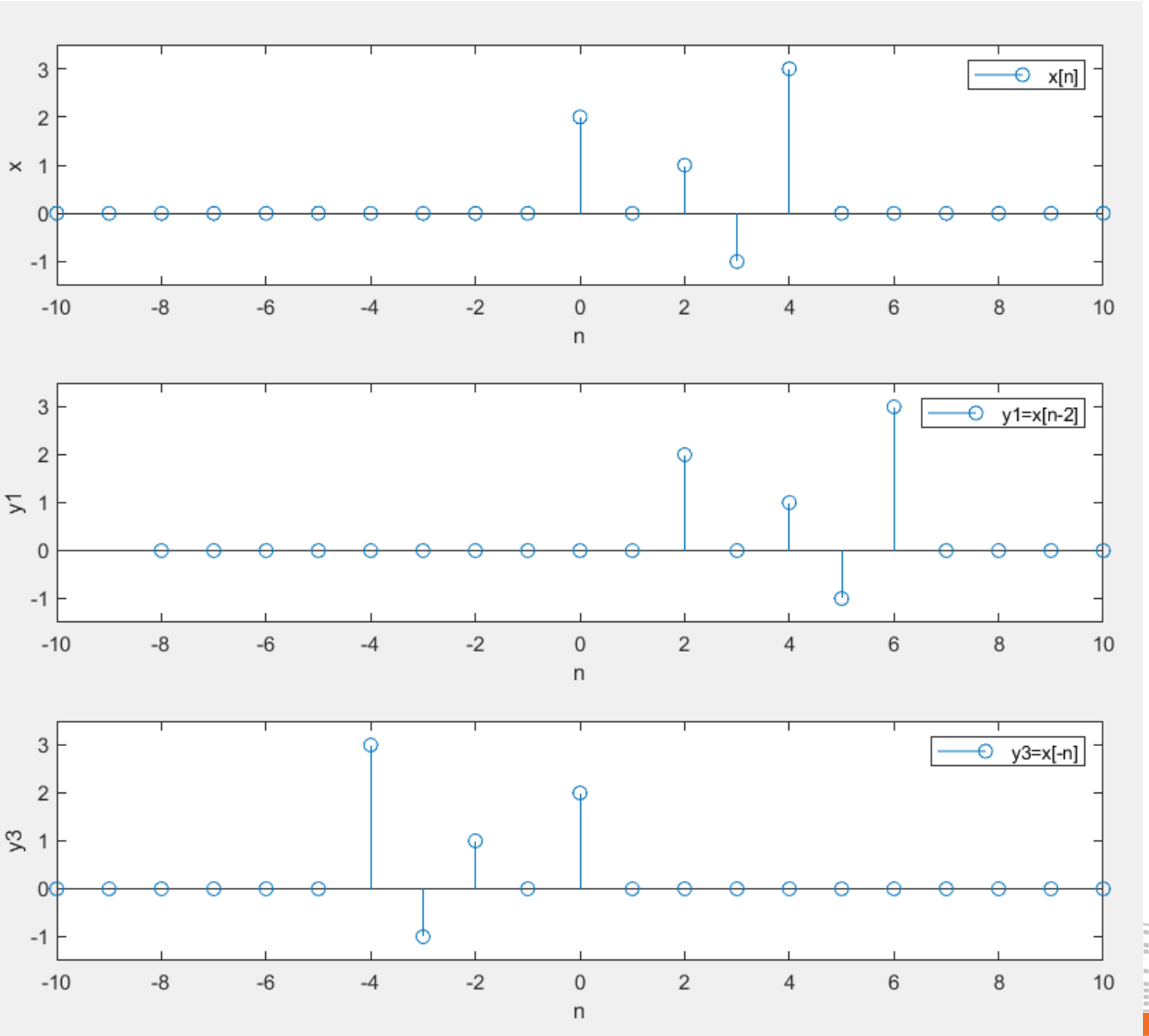
Obtained by flipping or shifting ?

$$y_1[n] = x[n-2],$$
$$y_2[n] = x[n+1],$$
$$y_3[n] = x[-n],$$
$$y_4[n] = x[-n+1].$$

```
% given nx, x

ny1 = nx + 2;
y1 = x;

ny3 = nx;
y3 = fliplr(x);
```



Signals Basic / Topics

SUSTech Southern University of Science and Technology

- 离散时间正弦信号，Discrete-Time Sinusoidal Signals
- 连续时间复指数信号，Continuous-Time Complex Exponential Signals
- 离线时间信号的时间变换，Transformation of the Time Index for Discrete-Time Signals
- **连续时间信号的时间变换，Transformations of the Time index for Continuous-Time Signals**
- 离散时间系统的性质，Properties of Discrete-Time Systems



Signals Basic / Transformation of the Time Index for CT Signals (a)

From Lab Book, 1.7)

Intermediate Problems

(a). Use `Heaviside` to define `f` to be a symbolic expression for $f(t)$ as specified in Eq. (1.10). Plot this symbolic expression using `ezplot`.

$$f(t) = t\big(u(t) - u(t-2)\big), \tag{1.10}$$

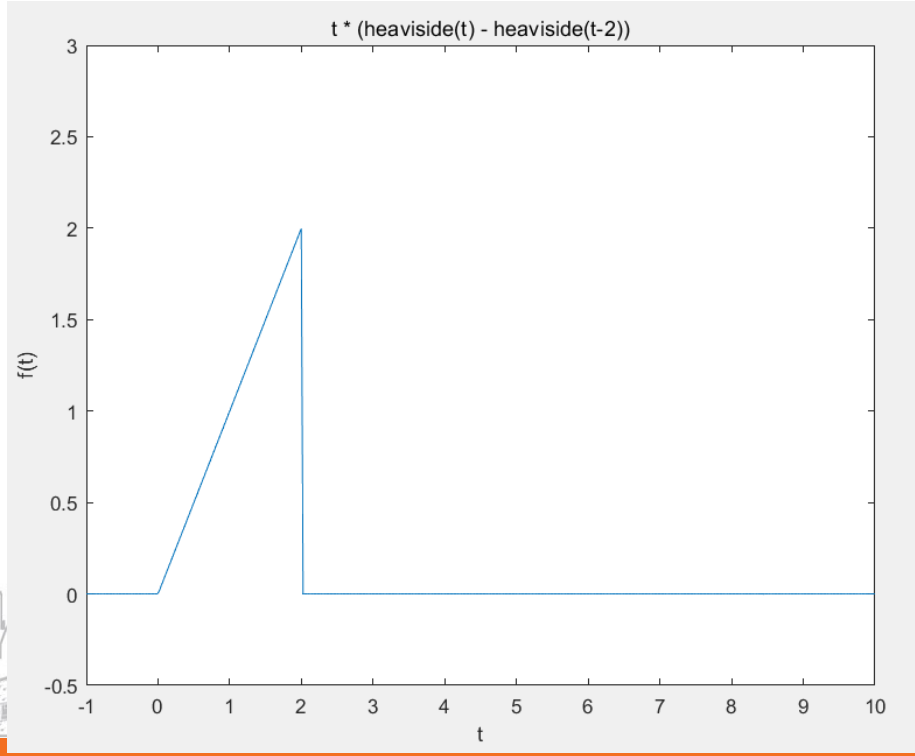
where $u(t)$ is the unit step signal

$$u(t) = \begin{cases} 1, & t \geq 0, \\ 0, & t < 0. \end{cases}$$

Function: heaviside

```
function f = Heaviside(t)
% HEAVISIDE Unit Step function
% f = Heaviside(t) returns a vector f the same size as
% the input vector, where each element of f is 1 if the
% corresponding element of t is greater than or equal to
% zero.
f = (t>=0);
```

```
4 -  syms t;
5 -  f = sym( t * (heaviside(t) - heaviside(t-2)) );
6
7 -  ezplot(f, [-1, 10]);
8 -  title("t * (heaviside(t) - heaviside(t-2))");
9 -  xlim([-1, 10]);
10 - ylim([-0.5, 3]);
11 - xlabel("t");
12 - ylabel("f(t)");
```



Signals Basic / Transformation of the Time Index for CT Signals (b, c)

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From Lab Book, 1.7)

(b). The expressions below define a set of continuous-time signals in terms of $f(t)$. For each of the following signals, state how you expect it to be related to $f(t)$, e.g., "delayed by 7," "flipped then advanced by 16":

$$g_1(t) = f(-t)\,,$$
$$g_2(t) = f(t+1)\,,$$
$$g_3(t) = f(t-3)\,,$$
$$g_4(t) = f(-t+1)\,,$$
$$g_5(t) = f(-2t+1)\,.$$

(c). Use the Symbolic Math Toolbox function **subs** and the symbolic expression **f** you defined in Part (a) to define symbolic expressions in MATLAB called **g1** through **g5** to represent the signals in Part (b). Plot each signal using **ezplot** and state whether or not the plot agrees with your prediction from Part (b).

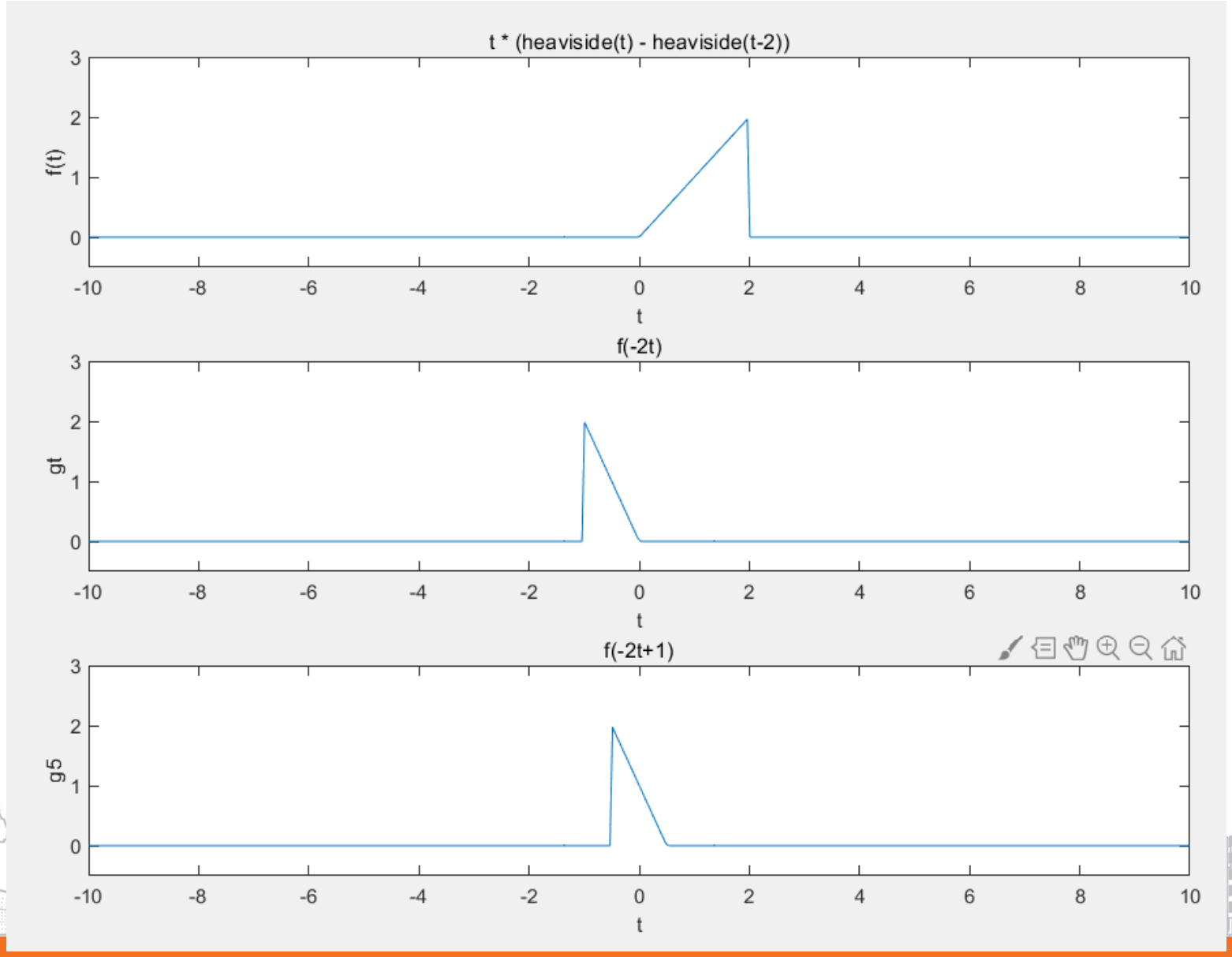
```
4 –     syms t;
5 –     f = sym( t * (heaviside(t) - heaviside(t-2)) );
6 –     subplot(3, 1, 1);
7 –     ezplot(f, [-10, 10]);
8 –     title("t * (heaviside(t) - heaviside(t-2))");
9 –     xlim([-10, 10]);
10 –    ylim([-0.5, 3]);
11 –    xlabel("t");
12 –    ylabel("f(t)");
13
14 –    gt = subs(f, t, -2*t);
15 –    subplot(3, 1, 2);
16 –    ezplot(gt, [-10, 10]);
17 –    title("f(-2t)");
18 –    xlim([-10, 10]);
19 –    ylim([-0.5, 3]);
20 –    xlabel("t");
21 –    ylabel("gt");
22
23 –    g5 = subs(f, t, -2*t + 1);
24 –    subplot(3, 1, 3);
25 –    ezplot(g5, [-10, 10]);
26 –    title("f(-2t+1)");
27 –    xlim([-10, 10]);
28 –    ylim([-0.5, 3]);
29 –    xlabel("t");
30 –    ylabel("g5");
```

Signals Basic / Transformation of the Time Index for CT Signals (b, c)

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After horizontal flipping,
Scale by 1/2 along the X-axis

Shift 1/2 units to the right



Signals Basic / Topics

SUSTech Southern University of Science and Technology

- 离散时间正弦信号，Discrete-Time Sinusoidal Signals
- 连续时间复指数信号，Continuous-Time Complex Exponential Signals
- 离线时间信号的时间变换，Transformation of the Time Index for Discrete-Time Signals
- 连续时间信号的时间变换，Transformations of the Time index for Continuous-Time Signals
- **离散时间系统的性质，Properties of Discrete-Time Systems**



Signals Basic / Properties of Discrete-Time Systems (a)

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System satisfies: linearity, causality, stability, invertibility ?

(a). The system $y[n] = \sin((\pi/2)x[n])$ is not linear. Use the signals $x_1[n] = \delta[n]$ and $x_2[n] = 2\delta[n]$ to demonstrate how the system violates linearity.

From Lab Book, 1.4)

```
4  n = -10:10

5

6  x1 = (n == 0);
7  x2 = 2 * (n == 0);
8  x3 = x1 + x2;

9

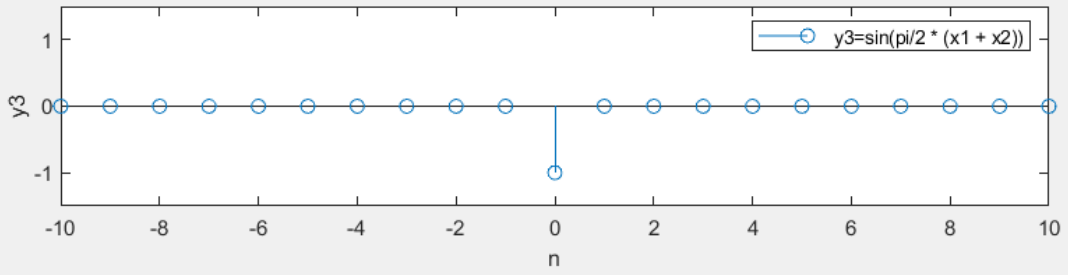
10 y1 = sin(pi / 2 * x1);
11 y2 = sin(pi / 2 * x2);

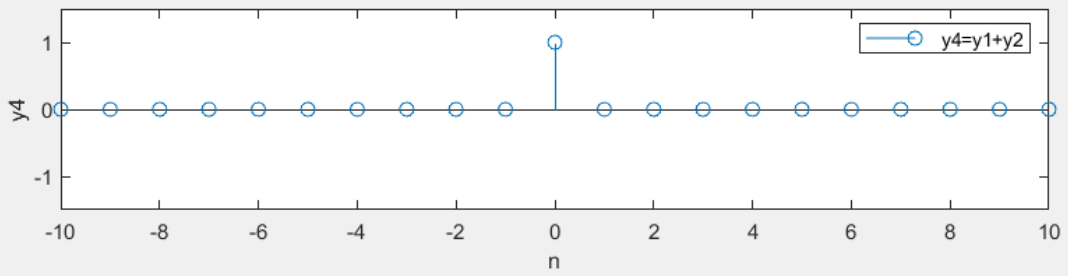
12

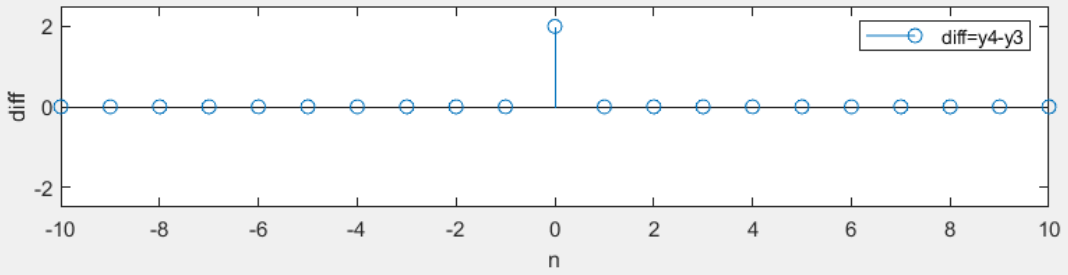
13 y3 = sin(pi / 2 * x3);
14 y4 = y1 + y2;

15

16 diff = y4 - y3;
```



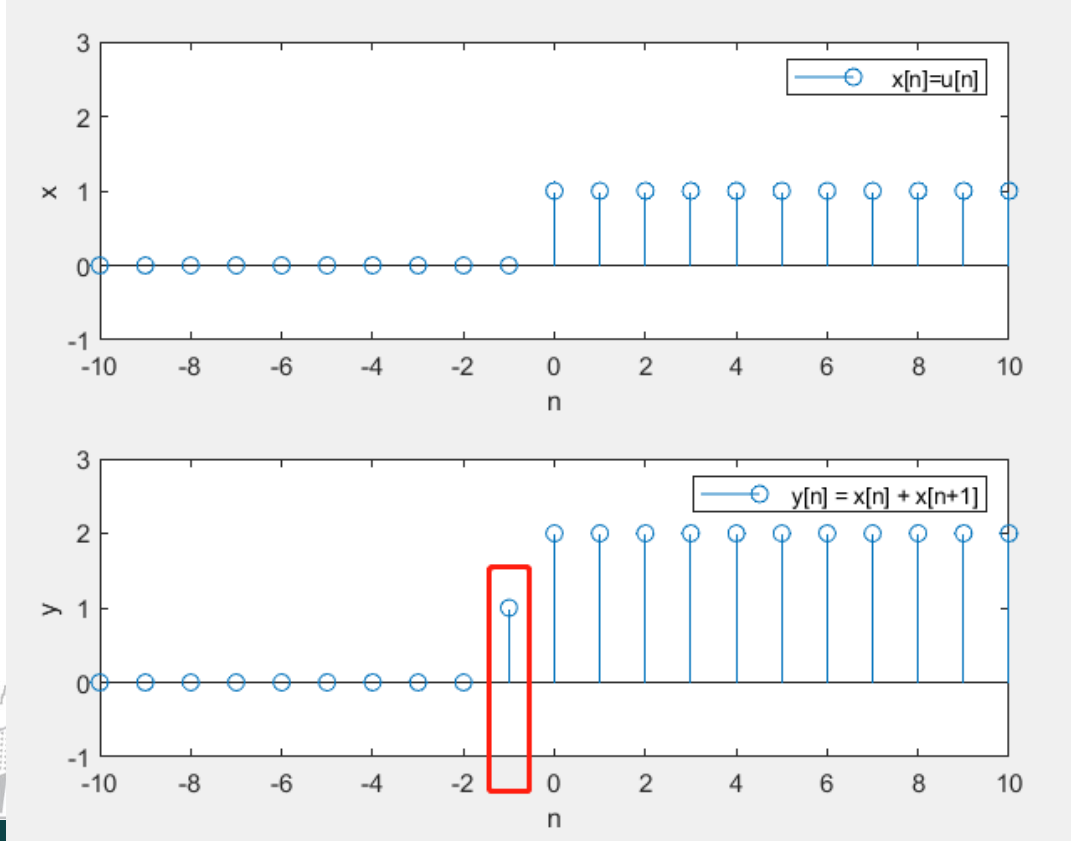




Signals Basic / Properties of Discrete-Time Systems (b)

System satisfies: linearity, causality, stability, invertibility ?

(b). The system $y[n] = x[n] + x[n+1]$ is not causal. Use the signal $x[n] = u[n]$ to demonstrate this. Define the MATLAB vectors x and y to represent the input on the interval $-5 \leq n \leq 9$, and the output on the interval $-6 \leq n \leq 9$, respectively.




```
4   n = -10:10
5
6   x = (n >= 0);
7   x_bw1 = ((n+1) >= 0);
8   y = x + x_bw1;
9
10  subplot(2, 1, 1);
11  stem(n, x);
12  ylim([-1, 3]);
13  xlabel("n");
14  ylabel("x");
15  legend(' x[n]=u[n]');
16
17  subplot(2, 1, 2);
18  stem(n, y);
19  ylim([-1, 3]);
20  xlabel("n");
21  ylabel("y");
22  legend('y[n] = x[n] + x[n+1]');
```

Signals Basic / Properties of Discrete-Time Systems (c)

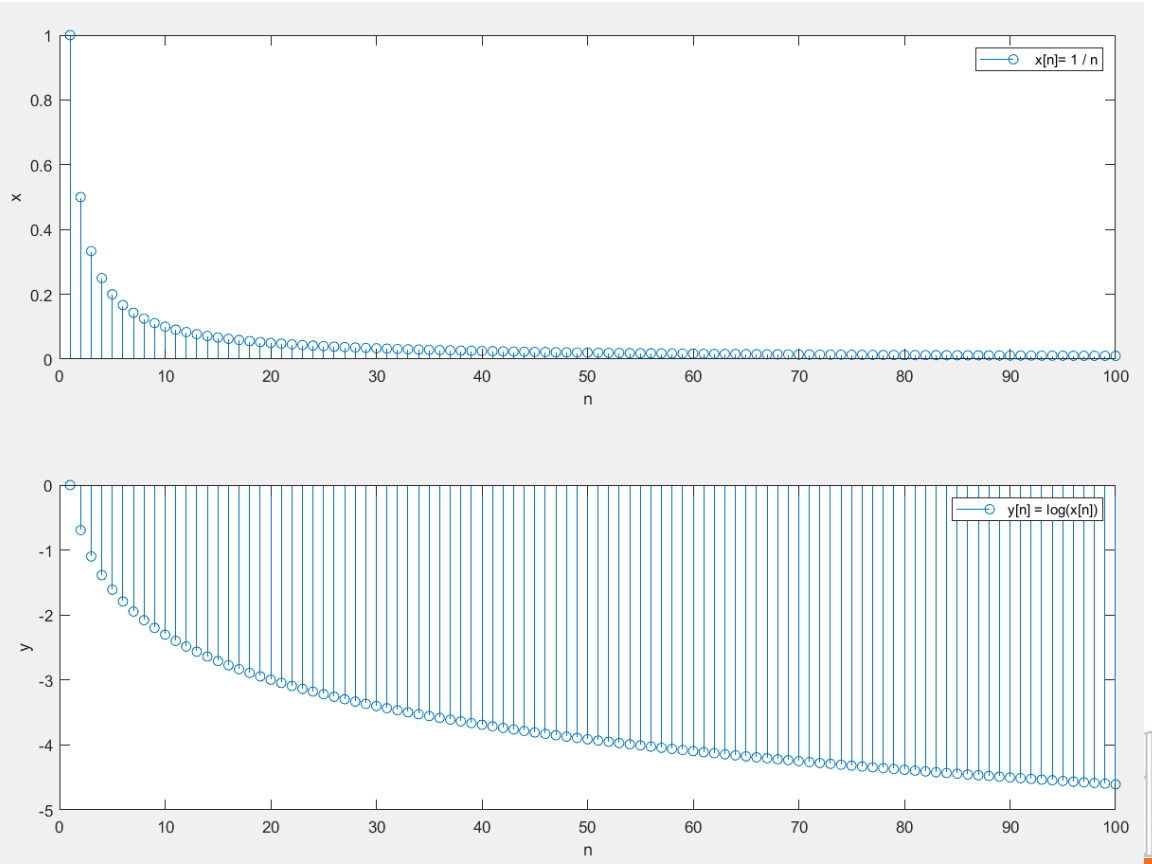
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System satisfies: linearity, causality, stability, invertibility ?

(c). The system $y[n] = \log(x[n])$ is not stable.

(d). The system given in Part (a) is not invertible.

```
4 -  n = 1:100
5 -  x = 1 ./ n;
6 -  y = log(x);
7
8 -  subplot(2, 1, 1);
9 -  stem(n, x);
10 - xlabel("n");
11 - ylabel("x");
12 - legend(' x[n]= 1 / n');
13
14 - subplot(2, 1, 2);
15 - stem(n, y);
16 - xlabel("n");
17 - ylabel("y");
18 - legend('y[n] = log(x[n])');
```



Signals Basic / Properties of Discrete-Time Systems (d)

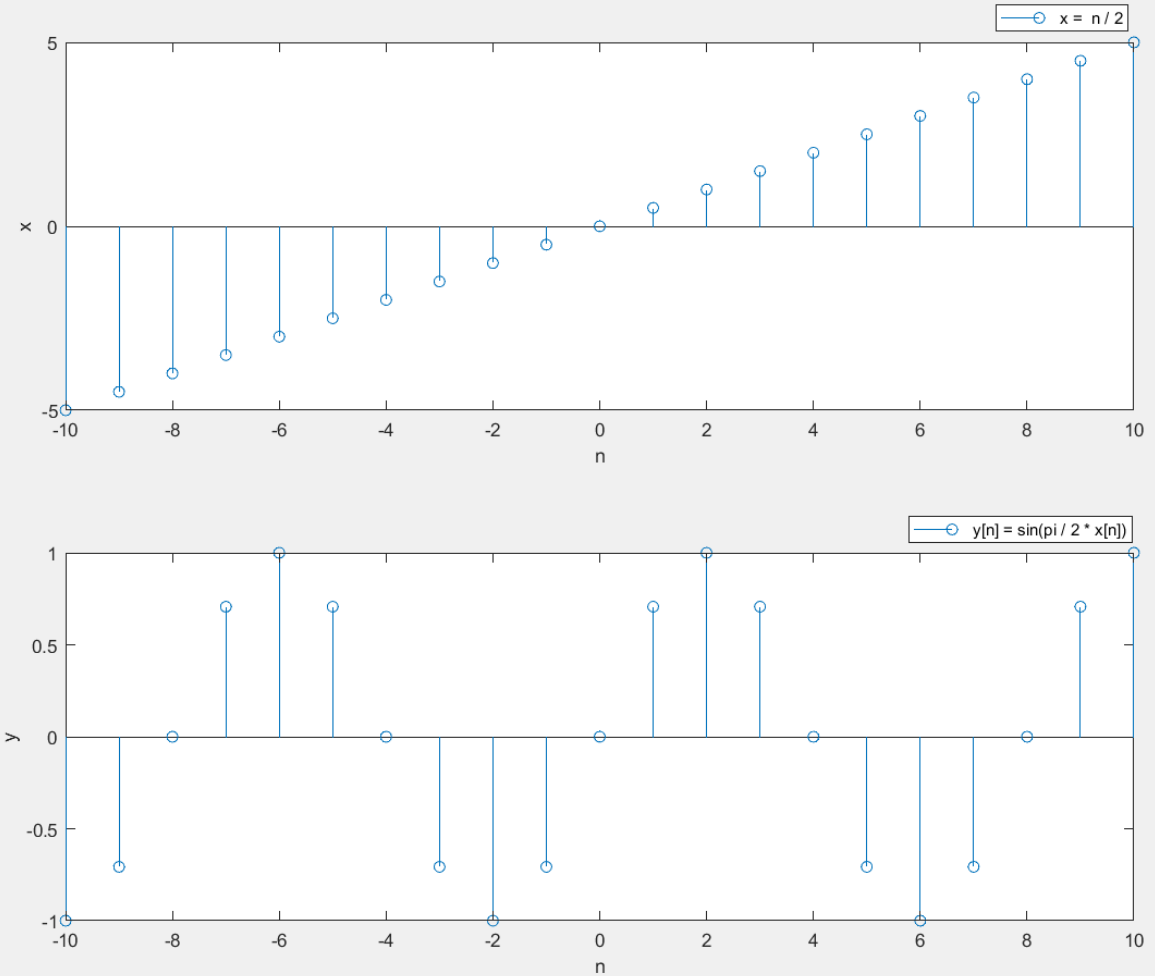
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System satisfies: linearity, causality, stability, invertibility ?

(c). The system $y[n] = \log(x[n])$ is not stable.

(d). The system given in Part (a) is not invertible.

```
4 –   n = -10:10
5
6 –   x = n / 2;
7 –   y = sin(pi / 2 * x);
8
9 –   subplot(2, 1, 1);
10 –  stem(n, x);
11 –  xlabel("n"); ylabel("x"); ylim([-5, 5]);
12 –  legend('x =  n / 2');
13
14 –  subplot(2, 1, 2);
15 –  stem(n, y);
16 –  xlabel("n"); ylabel("y"); ylim([-1, 1]);
17 –  legend('y[n] = sin(pi / 2 * x[n])');
```



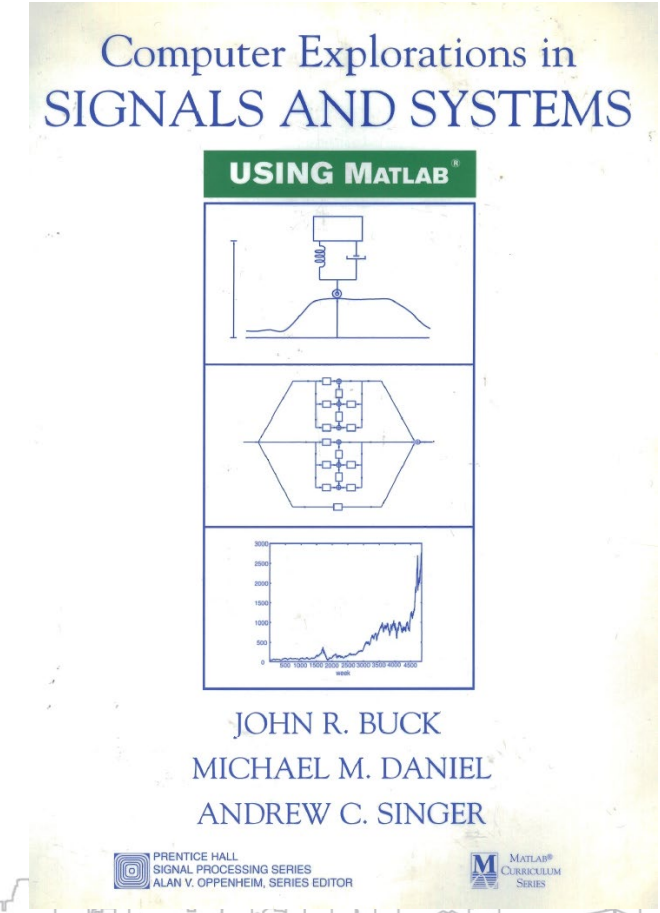
Lab1 / Task 2

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1.2, 1.3, 1.4: Next lab class

- 1.2: a); b); c) for x3 only; d) x3 only; e)
- 1.3: a); b) for y1, y4 only; c)
- 1.4: a); b); c); d); g)

1 Signals and Systems 1
1.1 Tutorial: Basic MATLAB Functions for Representing Signals 2
1.2 Discrete-Time Sinusoidal Signals 7
1.3 Transformations of the Time Index for Discrete-Time Signals 8
1.4 Properties of Discrete-Time Systems 10
1.5 Implementing a First-Order Difference Equation 11
1.6 Continuous-Time Complex Exponential Signals Ⓢ 12
1.7 Transformations of the Time Index for Continuous-Time Signals Ⓢ 14
1.8 Energy and Power for Continuous-Time Signals Ⓢ 16



Deadline: 2026/3/21 23:59

Content:

- Task 1(Optional)
- Task 2 (Must, follows the given template format)

Submit to Blackboard

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THANK YOU

